

CompTIA Linux+ (XK0-005)

Linux Identity Management

- Introduction
 - Lab Topology
 - Exercise 1 - Create and Configure User Accounts in Linux
 - Exercise 2 - Create and Configure Groups in Linux
 - Review
-

Introduction

XK0-005

User Accounts

Groups

Useradd

Groupadd

Welcome to the **Linux Identity Management** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Create and Configure User Accounts in Linux
- Exercise 2 - Create and Configure Groups in Linux

After completing this module, you should be able to:

- Create a User Account
- Modify the User Account
- Delete the User Account
- Create Groups
- Add Users to Groups

Exam Objectives

The following exam objectives are covered in this module:

2.2 Given a scenario, implement identity management

- Account creation and deletion
- Account management

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABSRV01** - Ubuntu 20.04 - Stand-Alone Ubuntu Server (Console)
- **PLABSRV02** - Alma Linux 8.5 - Stand-Alone Linux Server (GUI)
- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)
- **PLABWIN10** - Windows 10 - Stand-Alone Workstation

Click **Next** to proceed to the first exercise.

Exercise 1 - Create and Configure User Accounts in Linux

In this exercise, you will create and configure a user account on a Ubuntu device. User accounts on a Linux system can be managed through the Terminal console or the Graphical User Interface (GUI), depending on whether the GUI has been installed. The terminal is normally the preferred and time-efficient way of creating and managing user accounts.

Learning Outcomes

After completing this exercise, you should be able to:

- Create a User Account
- Modify the User Account
- Delete the User Account

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Create a User Account

In this task, you will create a user account using the terminal interface.

Step 1

Connect to **PLABCLIENT01**.

Login as **administrator** using the following **Password**:

Passw0rd

Press **Enter**.

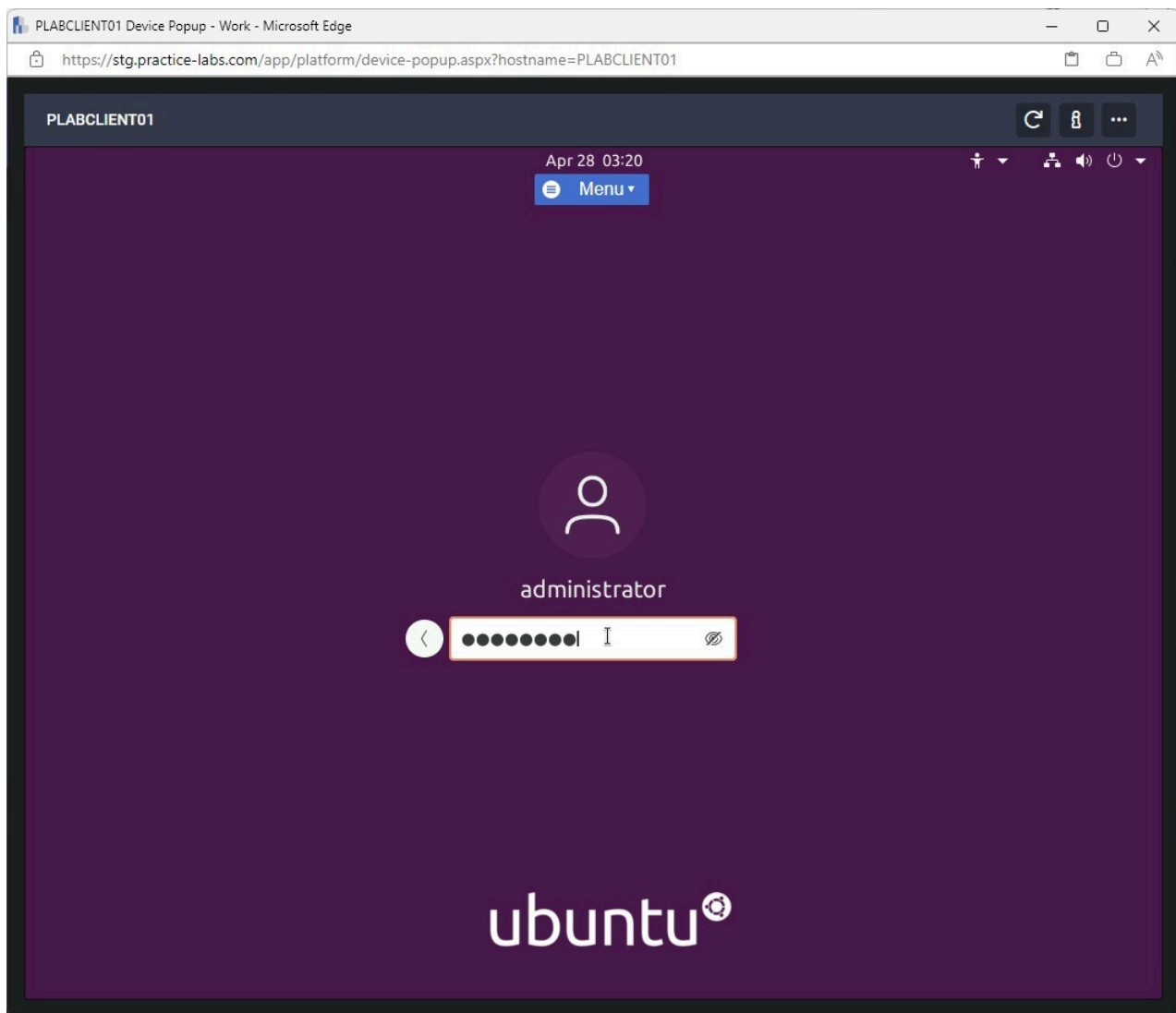


Figure 1.1 Screenshot of PLABCLIENT01: The login screen is displayed, showing the required credentials typed in.

Step 2

Click the **Show applications** icon on the bottom left corner.

Scroll down and select **Terminal**.

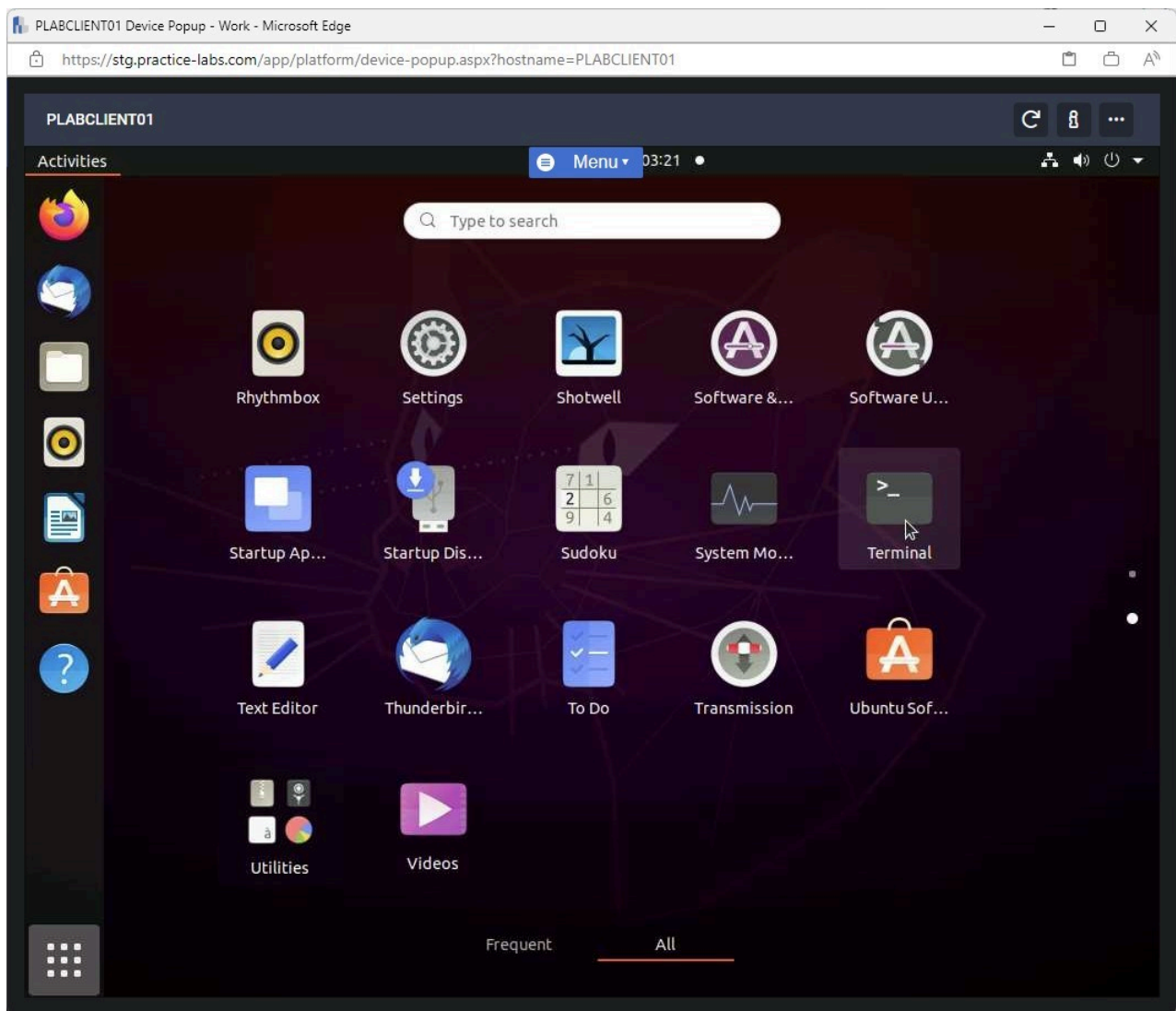


Figure 1.2 Screenshot of PLABCLIENT01: Show applications icon is clicked, and Terminal is selected.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
sudo useradd janedoe -c "Jane Doe" -s /bin/sh -d /home/janedoe -e 2026/12/31
```

Type the following **Password** and press **Enter**:

Passw0rd

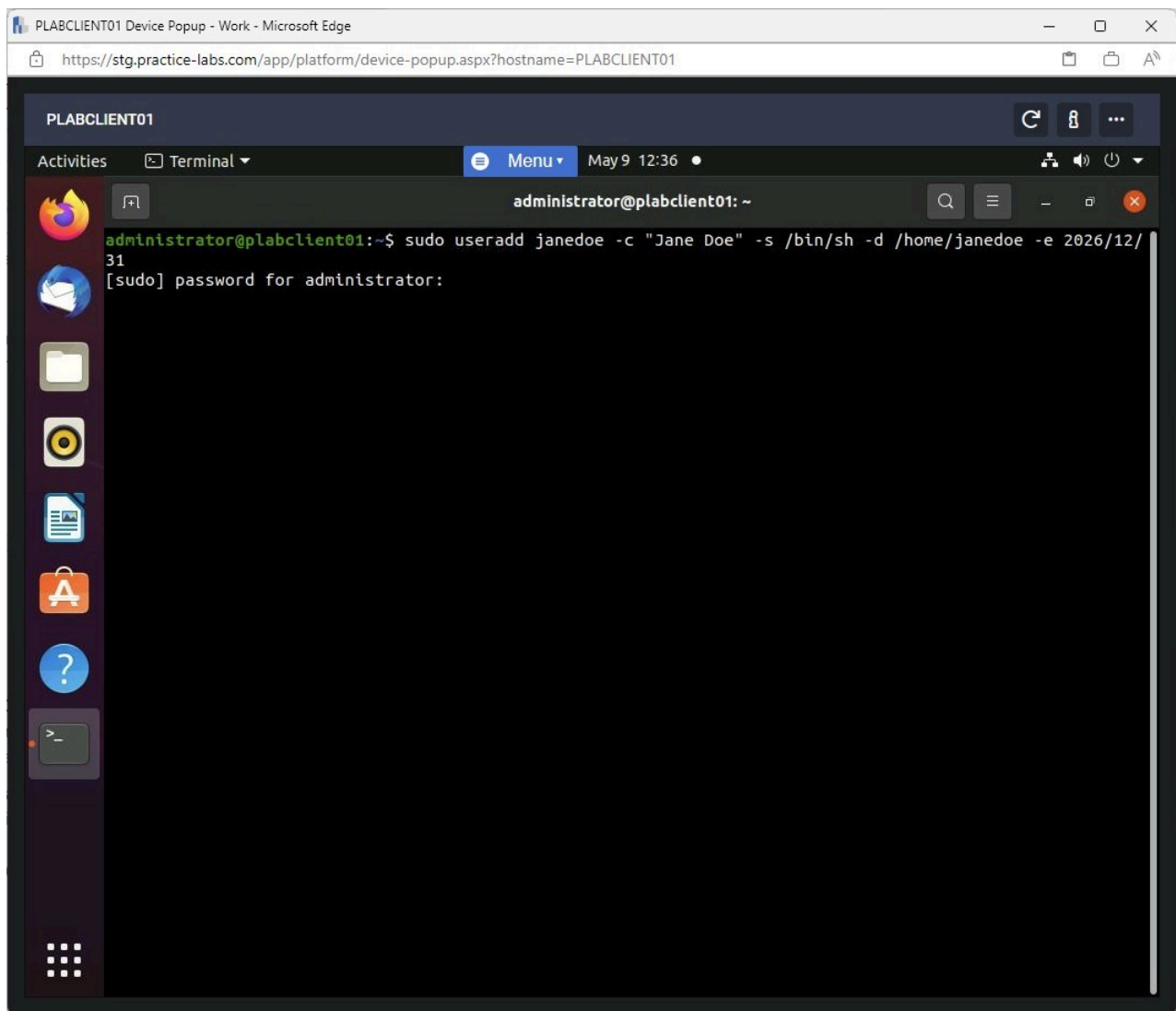


Figure 1.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The **useradd** command will create the **janedoe** account and sets the following:

- **-c** = display name, e.g., "Jane Doe"
- **-s** = default shell e.g., /bin/sh -bash shell
- **-d** = home directory e.g., /home/janedoe
- **-e** = expiry date, e.g., 2026/12/31

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
sudo useradd -D
```

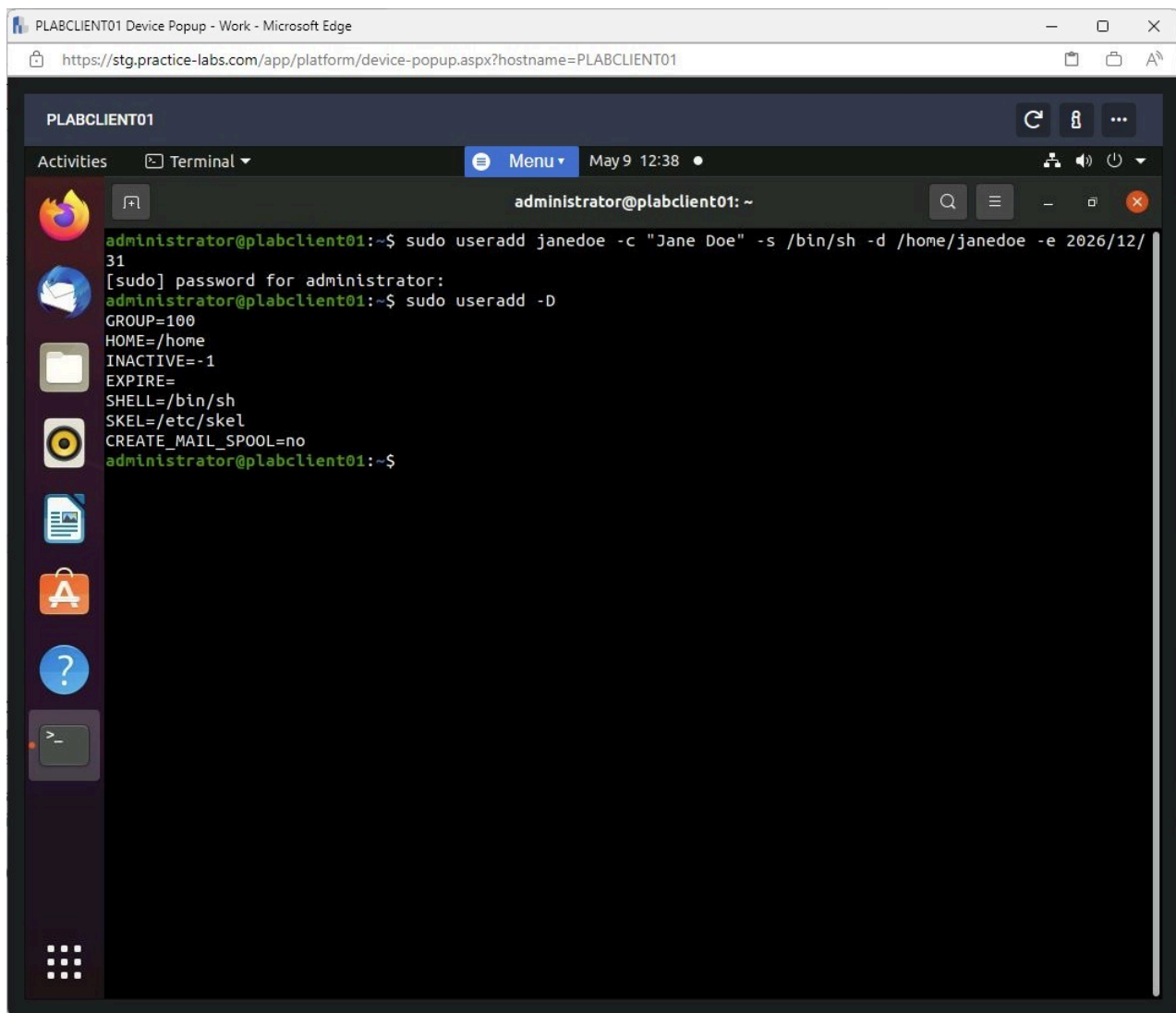


Figure 1.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **-D** option used with the **useradd** command will list the default values for **useradd**. These values are stored here:

- /etc/login.defs
- /etc/default/useradd
- /etc/skel

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
sudo useradd -m janedoe
```

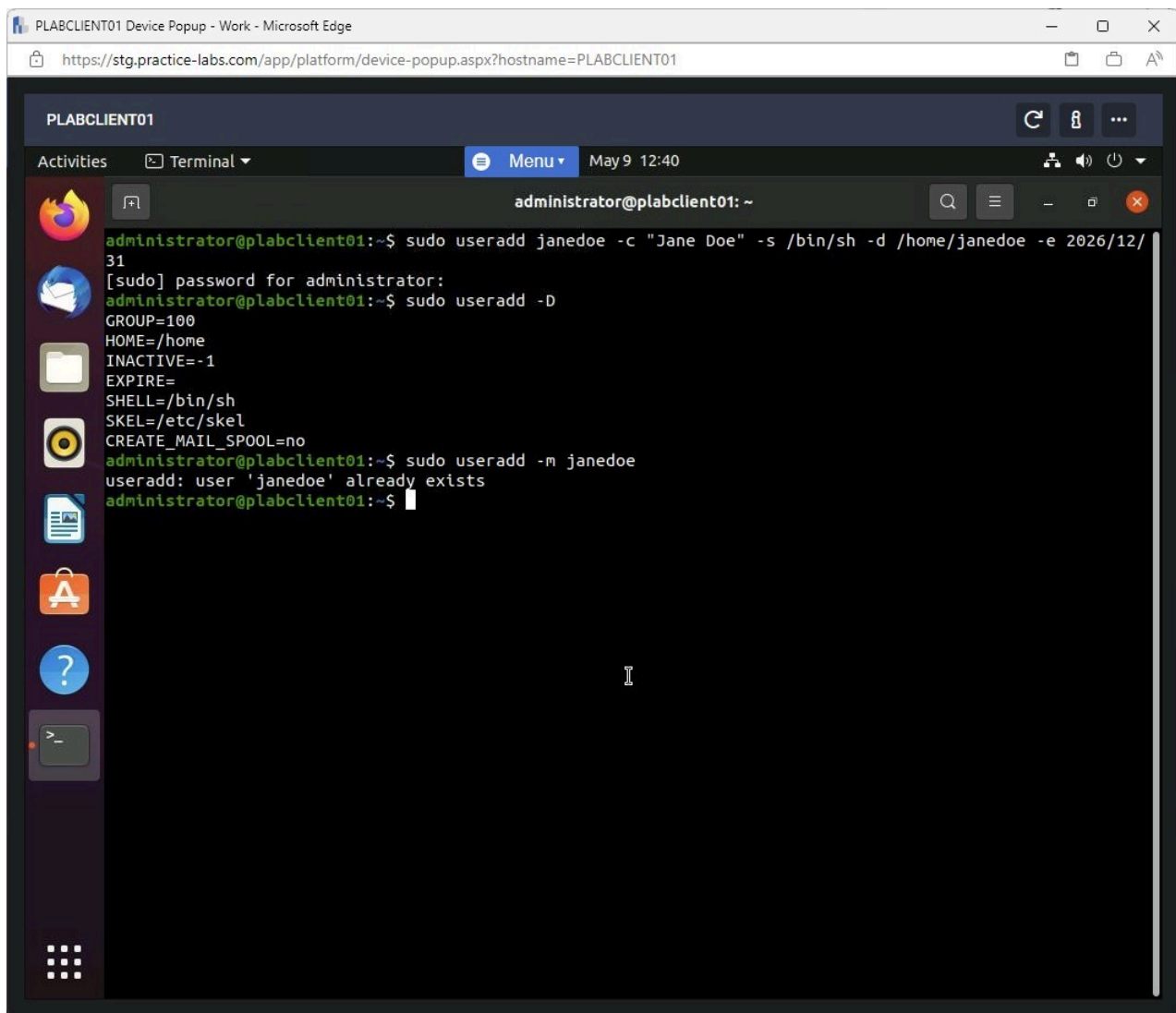


Figure 1.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **-m** option will create the new user's home directory with the default values listed in `/etc/defaults/useradd`.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
sudo passwd janedoe
```

For the **New password** prompt, type the following and press **Enter**:

Passw0rd

For the **Retype new password** prompt, type the following and press **Enter**:

Passw0rd

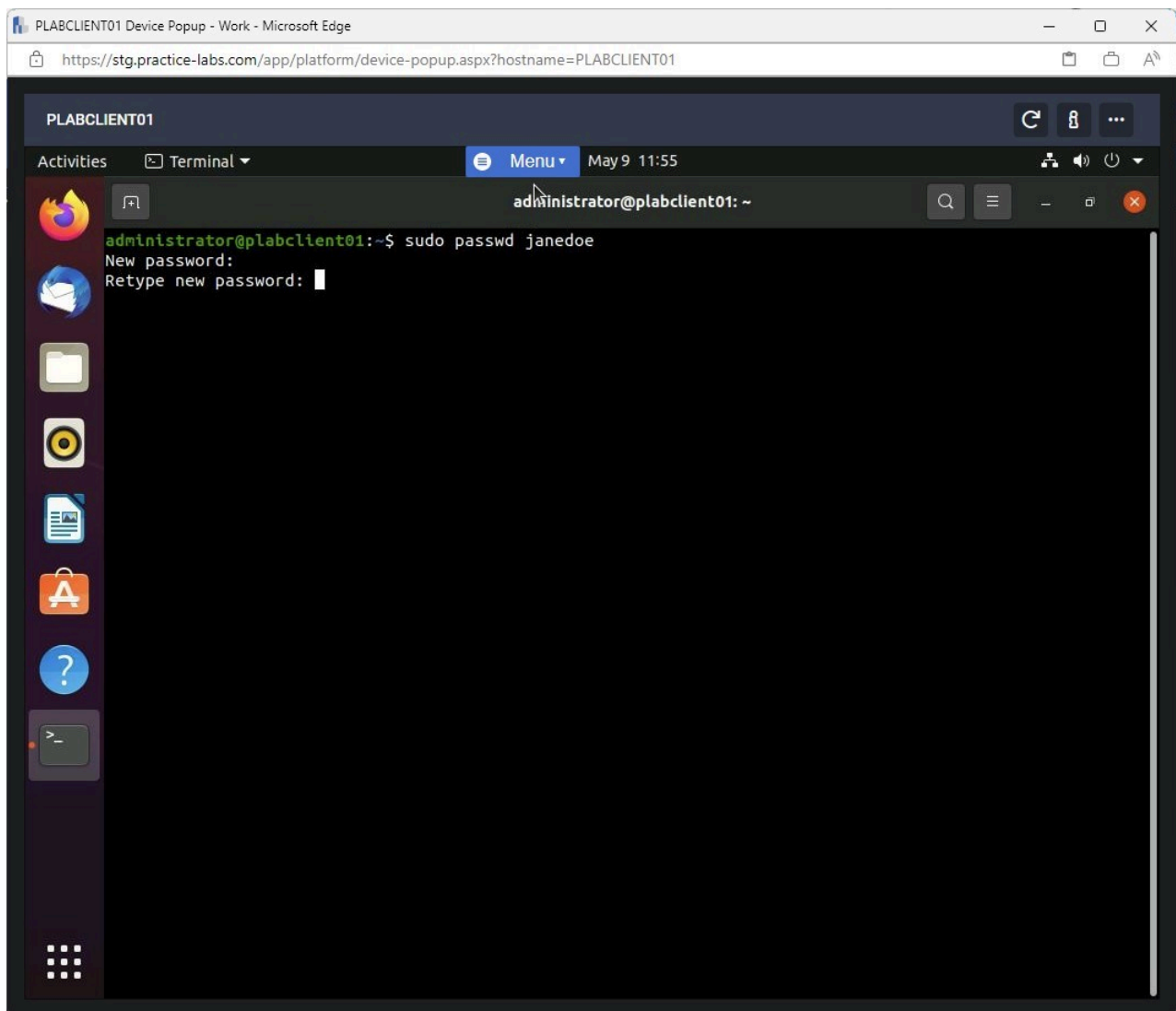


Figure 1.6 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 7

The password for **janedoe** is now set successfully using the **passwd** command.

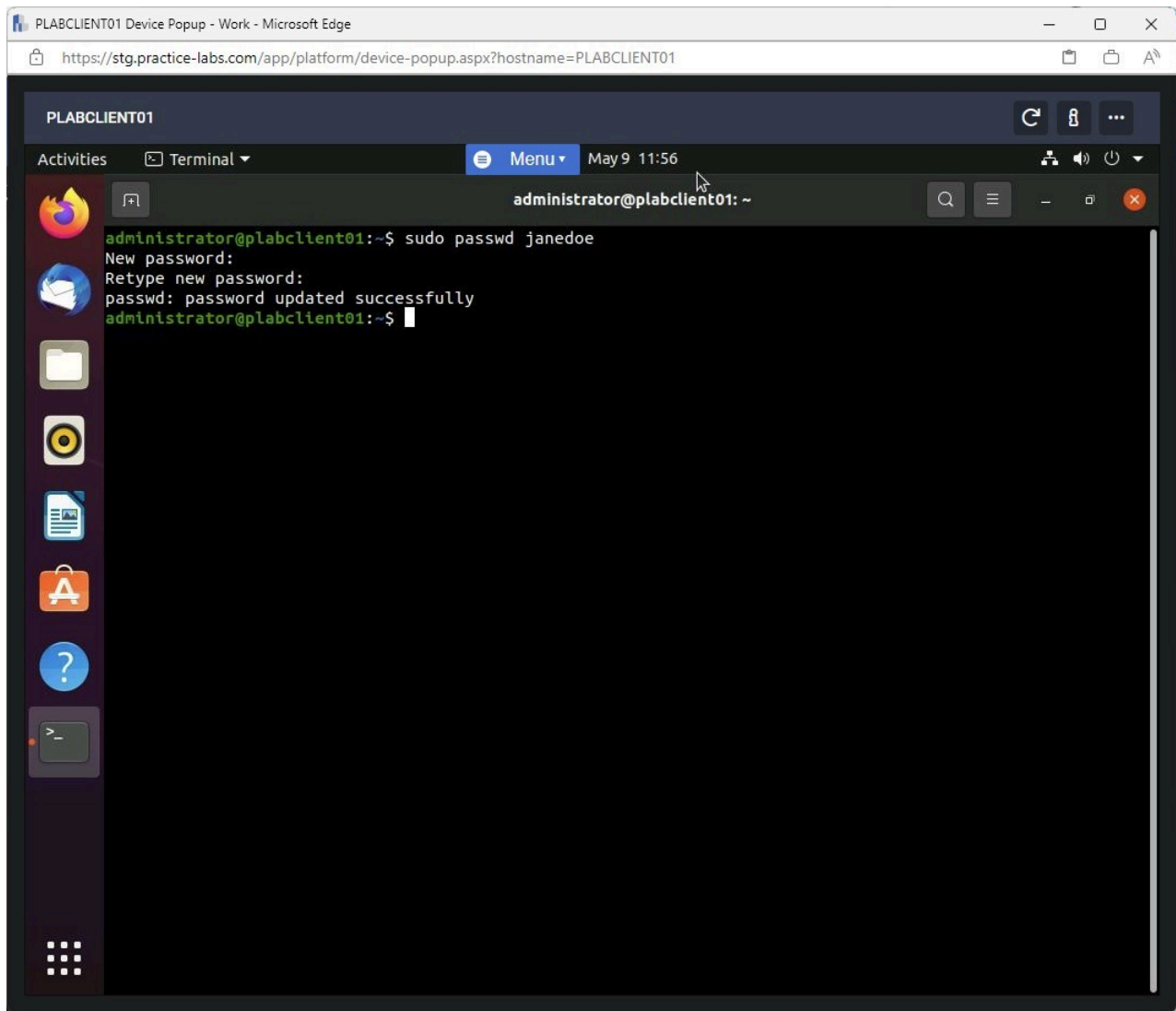


Figure 1.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the command output.

Step 8

Click the **Power** icon on the top right-hand corner and select **Power Off / Logout > Log Out** menu options from the drop-down menu.

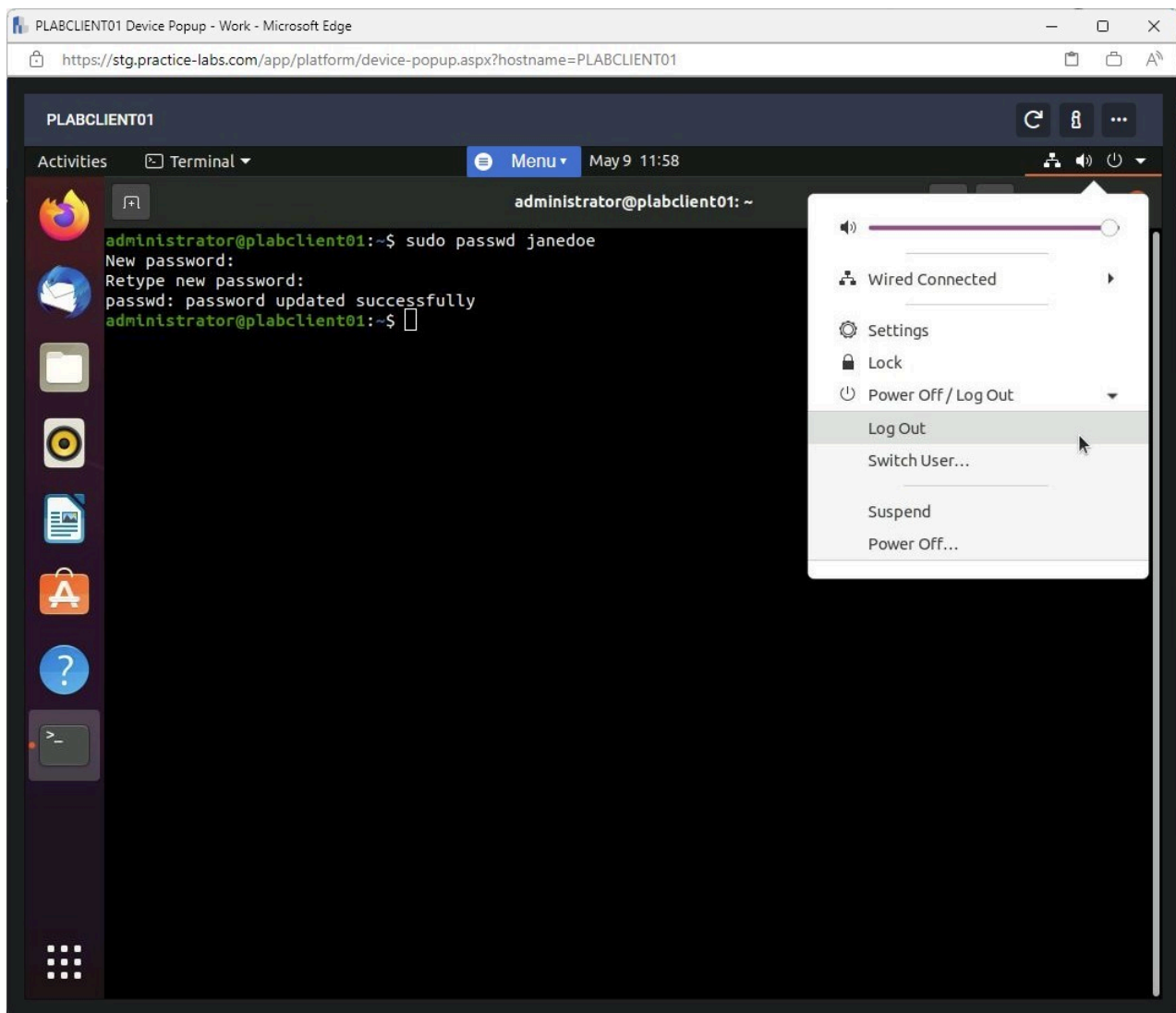


Figure 1.8 Screenshot of PLABCLIENT01: The Power icon is accessed, and Power Off / Logout > Log Out menu options are selected.

Step 9

In the **Log Out administrator** pop window, click **Log Out**.

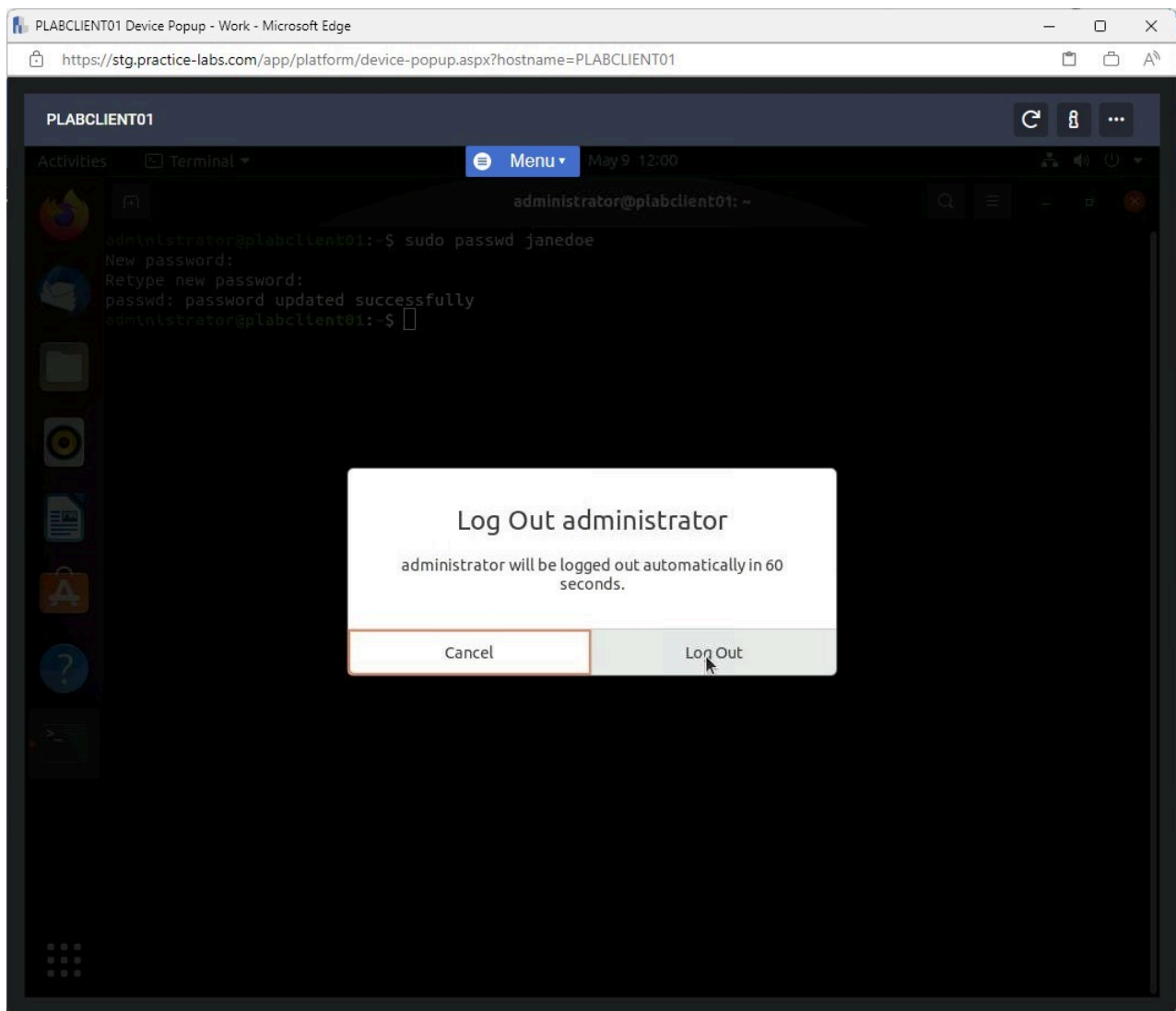


Figure 1.9 Screenshot of PLABCLIENT01: Log Out administrator pop window is displayed, showing Log Out selected.

Step 10

In the login screen, select **Jane Doe**.

Type the following **Password** and press **Enter**:

Passw0rd

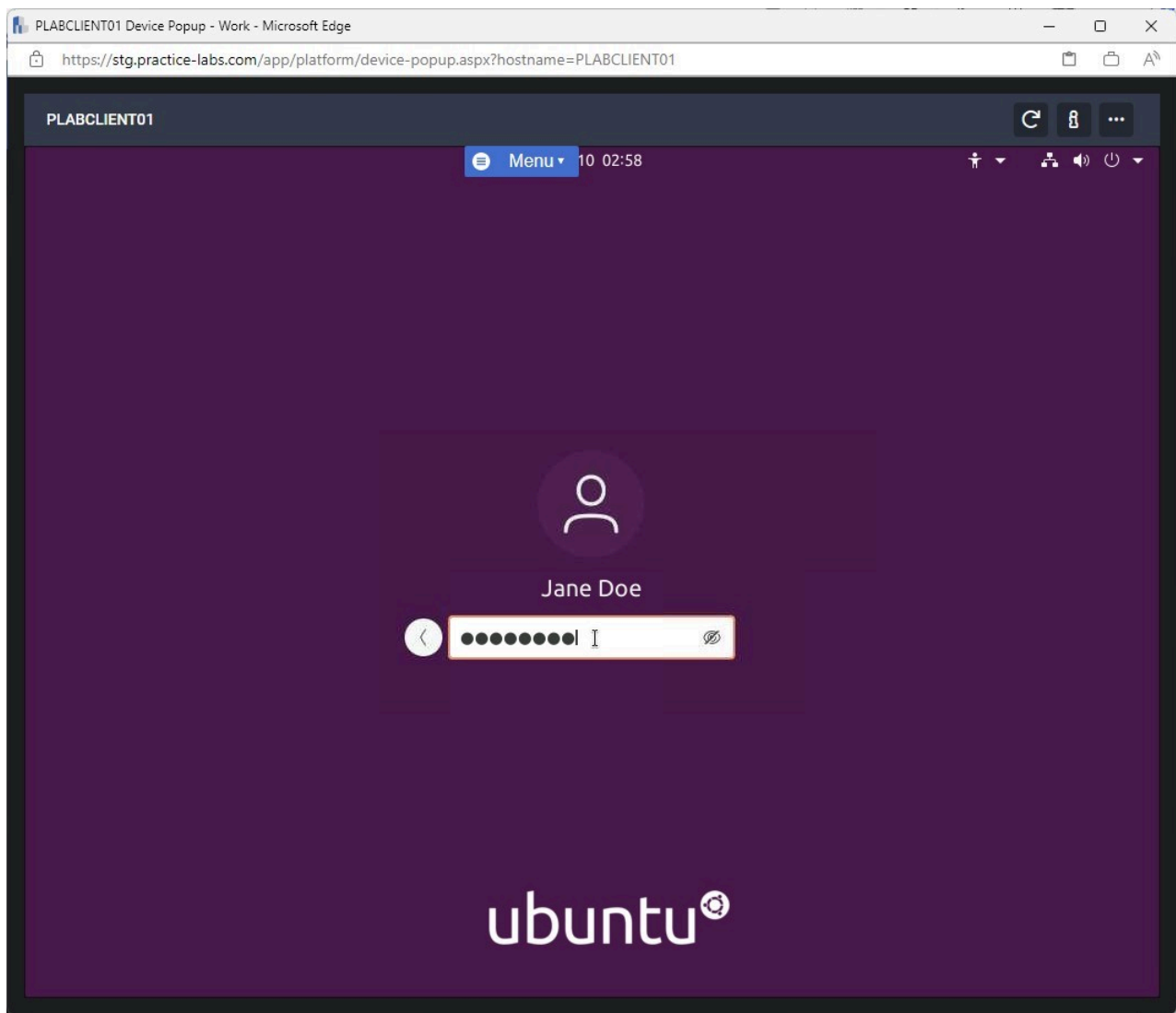


Figure 1.10 Screenshot of PLABCLIENT01: The login screen is displayed, showing the required credentials typed in.

Step 11

You are now successfully able to log in with the new user account.

Click the **Power** icon on the top right-hand corner and select **Power Off / Logout > Log Out** menu options from the drop-down menu.

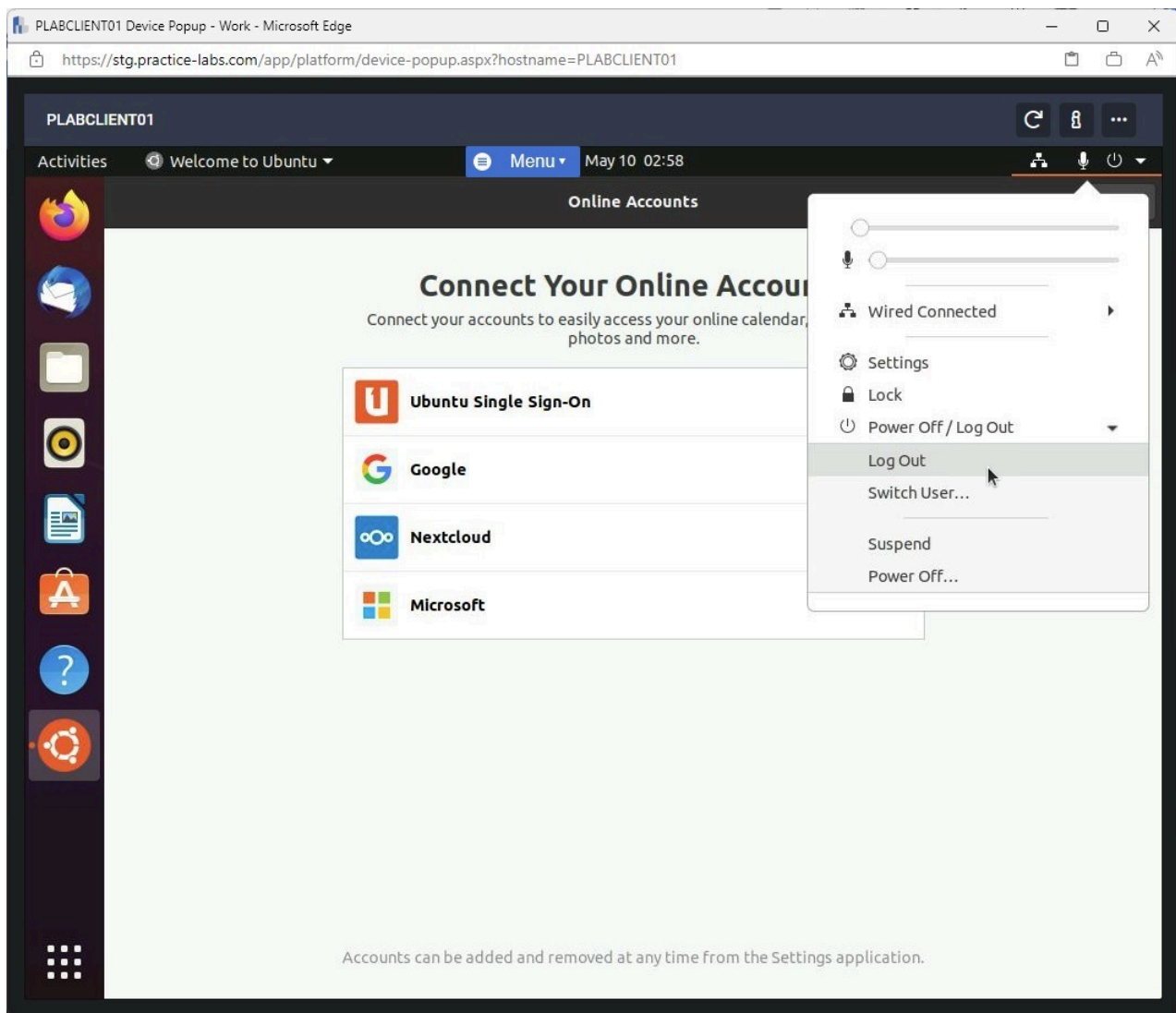


Figure 1.11 Screenshot of PLABCLIENT01: The Power icon is accessed, and Power Off / Log Out > Log Out menu options are selected.

Step 12

In the **Log Out Jane Doe** pop window, click **Log Out**.

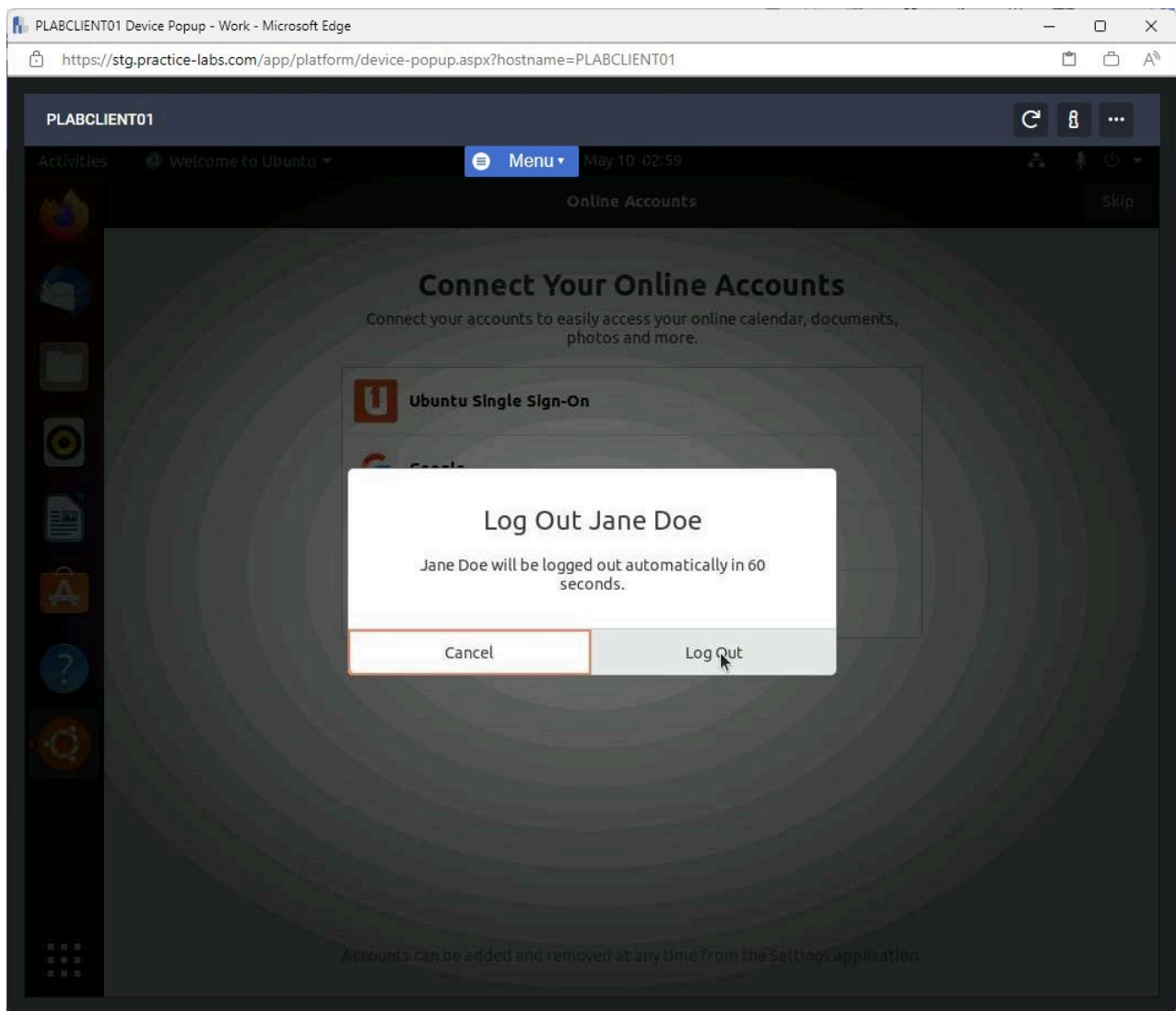


Figure 1.12 Screenshot of PLABCLIENT01: Log Out Jane Doe pop window is displayed, showing Log Out selected.

Step 13

Log back in as **administrator** using the password **Passw0rd**.

Click the **Show applications** icon on the bottom left corner. Scroll down and select **Terminal**.

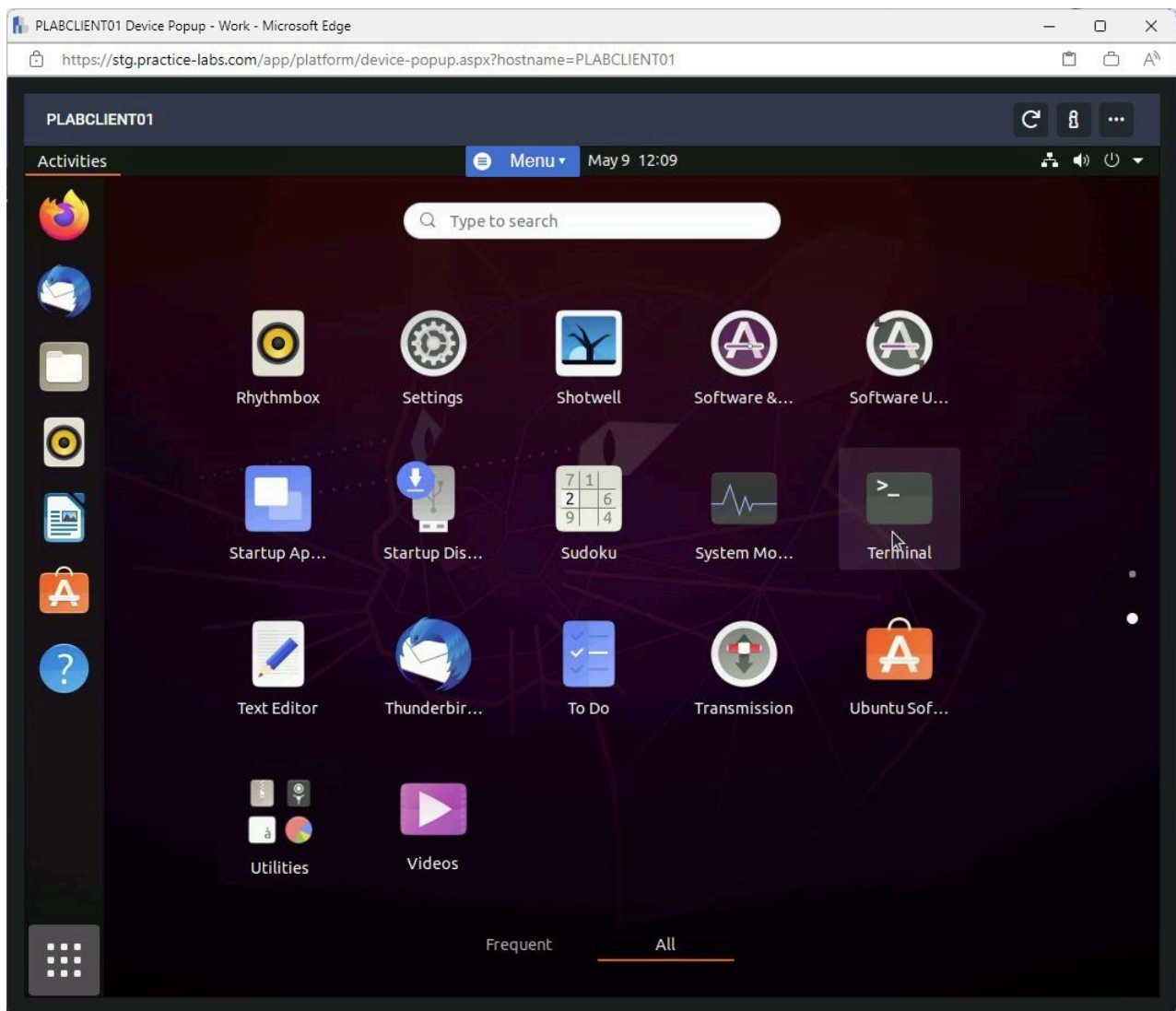


Figure 1.13 Screenshot of PLABCLIENT01: Show applications icon is clicked, and Terminal is selected.

Step 14

In the **Terminal** window, type the following command and press **Enter**:

```
sudo less /etc/passwd
```

Type the following **Password** and press **Enter**:

Passw0rd

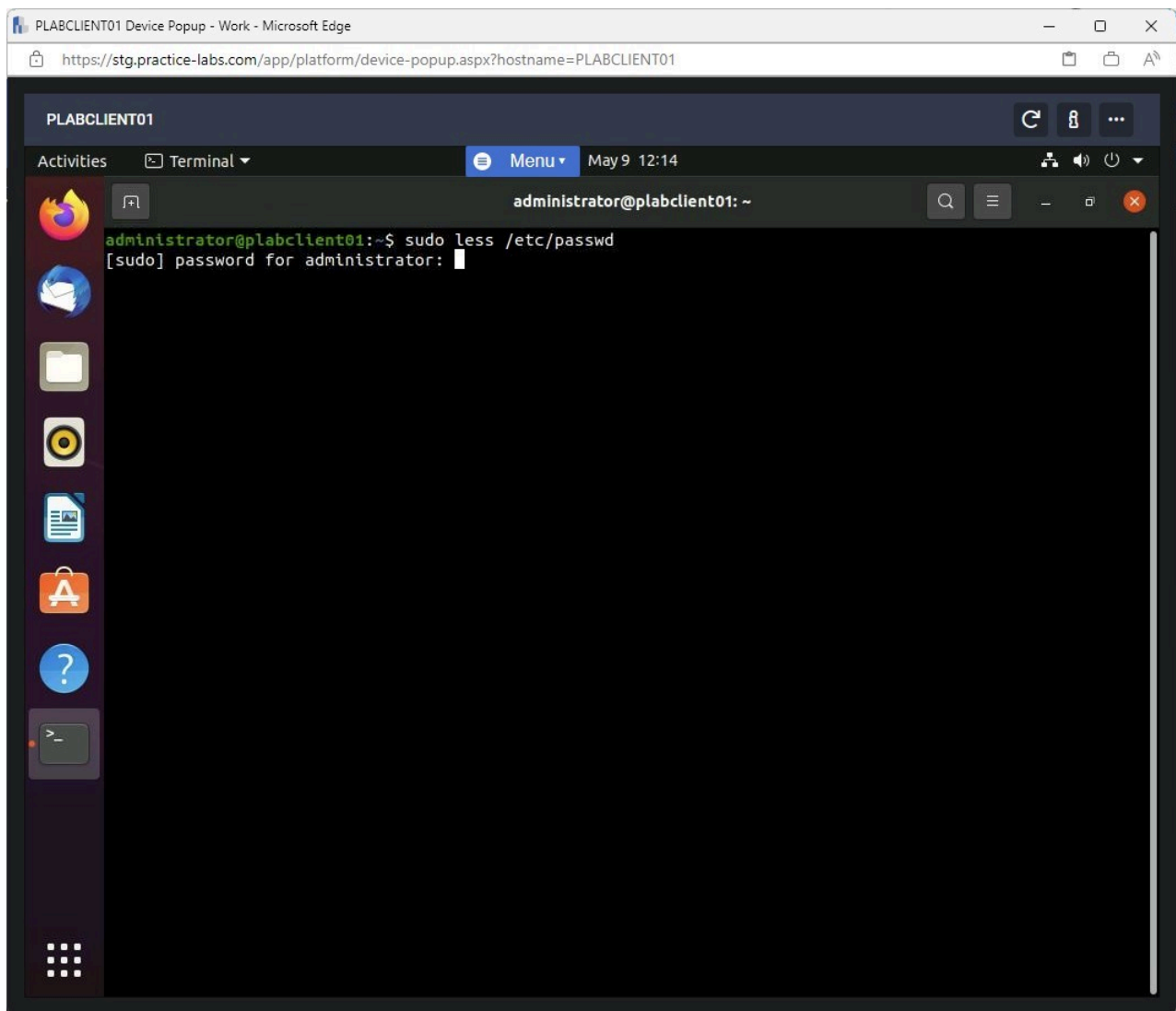


Figure 1.14 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The *less* command will open the */etc/passwd* text file, which contains a list of all the system accounts.

Step 15

Scroll down the text file. Notice the newly created **janedoe** account is listed.

```

PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 10 03:00
administrator@plabclient01: ~
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106:/nonexistent:/usr/sbin/nologin
syslog:x:104:110:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,:/var/lib/tpm:/bin/false
uidd:x:107:114:/run/uidd:/usr/sbin/nologin
tcpdump:x:108:115:/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:109:116:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:110:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin
rtkit:x:111:117:RealtimeKit,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:113:120:user for cups-pk-helper service,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:114:29:Speech Dispatcher,,:/run/speech-dispatcher:/bin/false
avahi:x:115:121:Avahi mDNS daemon,,:/var/run/avahi-daemon:/usr/sbin/nologin
kernoops:x:116:65534:Kernel Oops Tracking Daemon,,:/usr/sbin/nologin
saned:x:117:123:/var/lib/saned:/usr/sbin/nologin
nm-openvpn:x:118:124:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
hplip:x:119:7:HPLIP system user,,:/run/hplip:/bin/false
whoopsie:x:120:125:/nonexistent:/bin/false
colord:x:121:126:colord colour management daemon,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:122:127:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:123:128:PulseAudio daemon,,:/var/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:124:65534:/run/gnome-initial-setup:/bin/false
gdm:x:125:130:Gnome Display Manager:/var/lib/gdm3:/bin/false
sssd:x:126:131:SSSD system user,,:/var/lib/sss:/usr/sbin/nologin
administrator:x:1000:1000:administrator,,:/home/administrator:/bin/bash
systemd-coredump:x:999:999:systemd Core Dumper:/usr/sbin/nologin
janedoe:x:1001:1001:Jane Doe:/home/janedoe:/bin/sh
(END)

```

Figure 1.15 Screenshot of PLABCLIENT01: The /etc/passwd text file is displayed, showing the newly created account.

Step 16

Type the following to exit out of the window:

q

Back on the **Terminal** window, type the following command and press **Enter**:

sudo less /etc/shadow

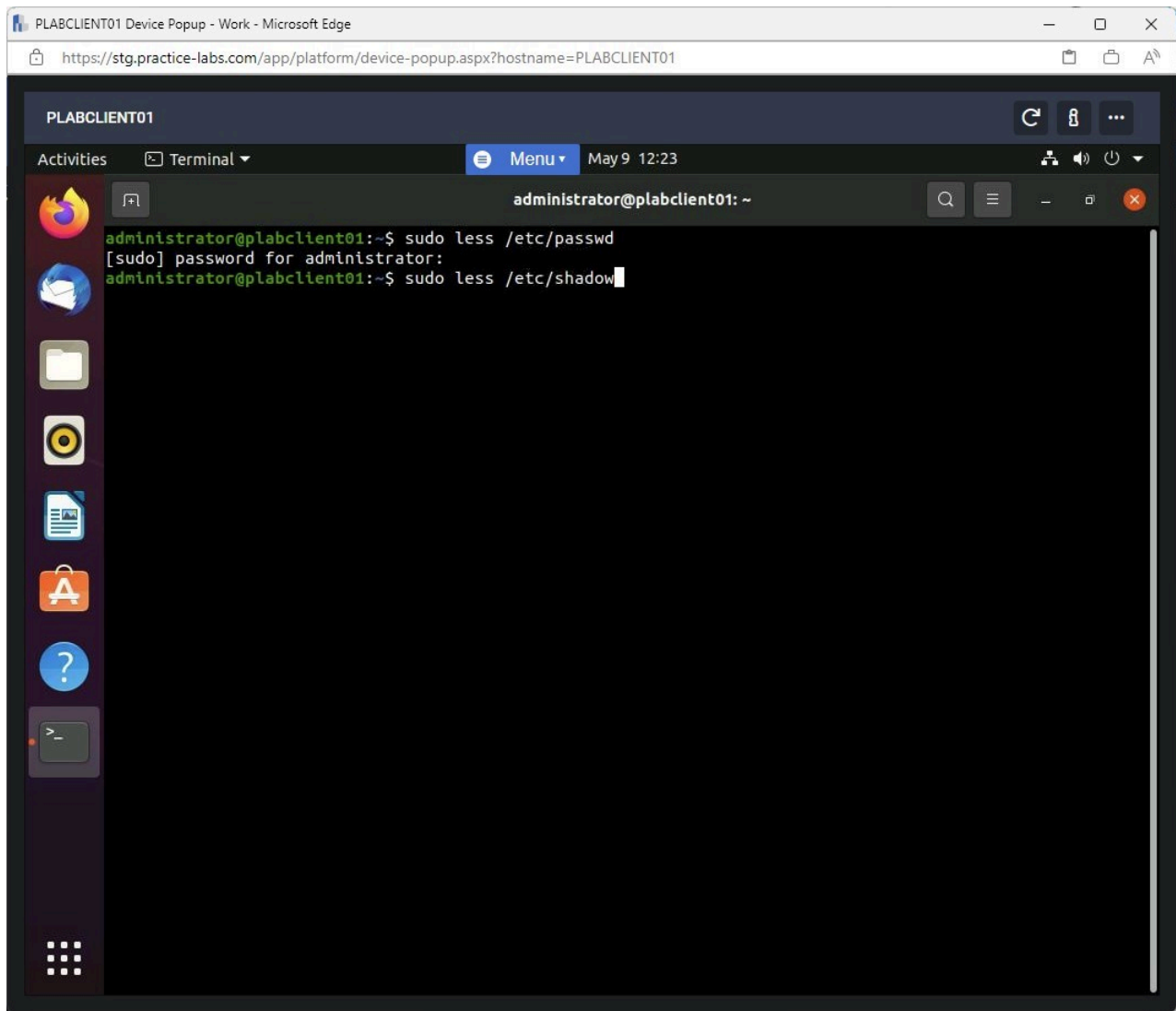


Figure 1.16 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The *less* command will open the */etc/shadow* text file, which contains a list of all the system accounts and the password hashed value.

Step 17

Scroll down until you see the **janedoe** account listed with its password hashed value.

Type the following to exit out of the window:

q

```

PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 10 03:01
administrator@plabclient01: ~

backup:*:19046:0:99999:7:::
list:*:19046:0:99999:7:::
irc:*:19046:0:99999:7:::
gnats:*:19046:0:99999:7:::
nobody:*:19046:0:99999:7:::
systemd-network:*:19046:0:99999:7:::
systemd-resolve:*:19046:0:99999:7:::
systemd-timesync:*:19046:0:99999:7:::
messagebus:*:19046:0:99999:7:::
syslog:*:19046:0:99999:7:::
_apt:*:19046:0:99999:7:::
tss:*:19046:0:99999:7:::
uidd:*:19046:0:99999:7:::
tcpdump:*:19046:0:99999:7:::
avahi-autoipd:*:19046:0:99999:7:::
usbmux:*:19046:0:99999:7:::
rtkit:*:19046:0:99999:7:::
dnsmasq:*:19046:0:99999:7:::
cups-pk-helper:*:19046:0:99999:7:::
speech-dispatcher:!:19046:0:99999:7:::
avahi:*:19046:0:99999:7:::
kernoops:*:19046:0:99999:7:::
saned:*:19046:0:99999:7:::
nm-openvpn:*:19046:0:99999:7:::
hplip:*:19046:0:99999:7:::
whoopsie:*:19046:0:99999:7:::
colord:*:19046:0:99999:7:::
geoclue:*:19046:0:99999:7:::
pulse:*:19046:0:99999:7:::
gnome-initial-setup:*:19046:0:99999:7:::
gdm:*:19046:0:99999:7:::
sssd:*:19046:0:99999:7:::
administrator:$6$xoAR0BGMPfsgwp.N$2ERFep/8PW50lmPE0dSKYQf.cTgnLCKwCqRLWs.aqCAww/.f3XcD5lXHAm860.QwN1Tvn
zsgo.wZ0EZoeHEw.:19072:0:99999:7:::
systemd-coredump:!:19072:0:99999:7:::
janedoe:$6$qsEVsFjWU1L1ZrUb$SaRiXCmVoTotHKr1d9FzSFuokcN20xk7e7K0ltC1T3r1t00DqIn7LfxeT4K6zFrqtUxpWdGqhY6B
RDwQdETyB0:19487:0:99999:7::20818:
(END)

```

Figure 1.17 Screenshot of PLABCLIENT01: The /etc/shadow text file is displayed, showing the newly created account with its password hashed value.

Step 18

In the **Terminal** window, type the following command and press **Enter**:

```
clear
```

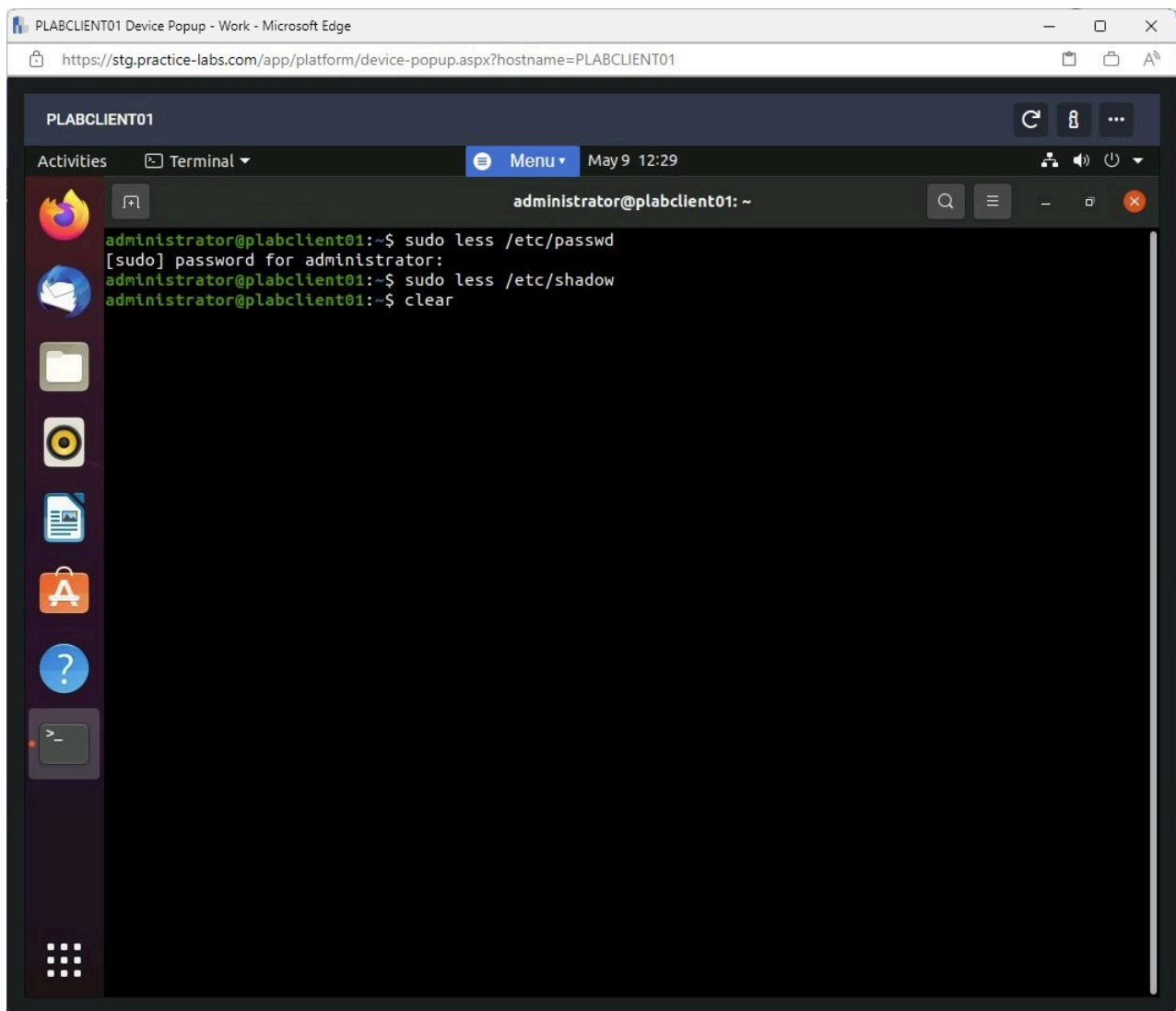


Figure 1.18 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Keep the **Terminal** window open.

Task 2 - Modify the User Account

In this task, you will use the **usermod** command to modify the user account.

Step 1

Ensure you are connected to **PLABCLIENT01**, where the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo usermod -l joedoe janedoe
```

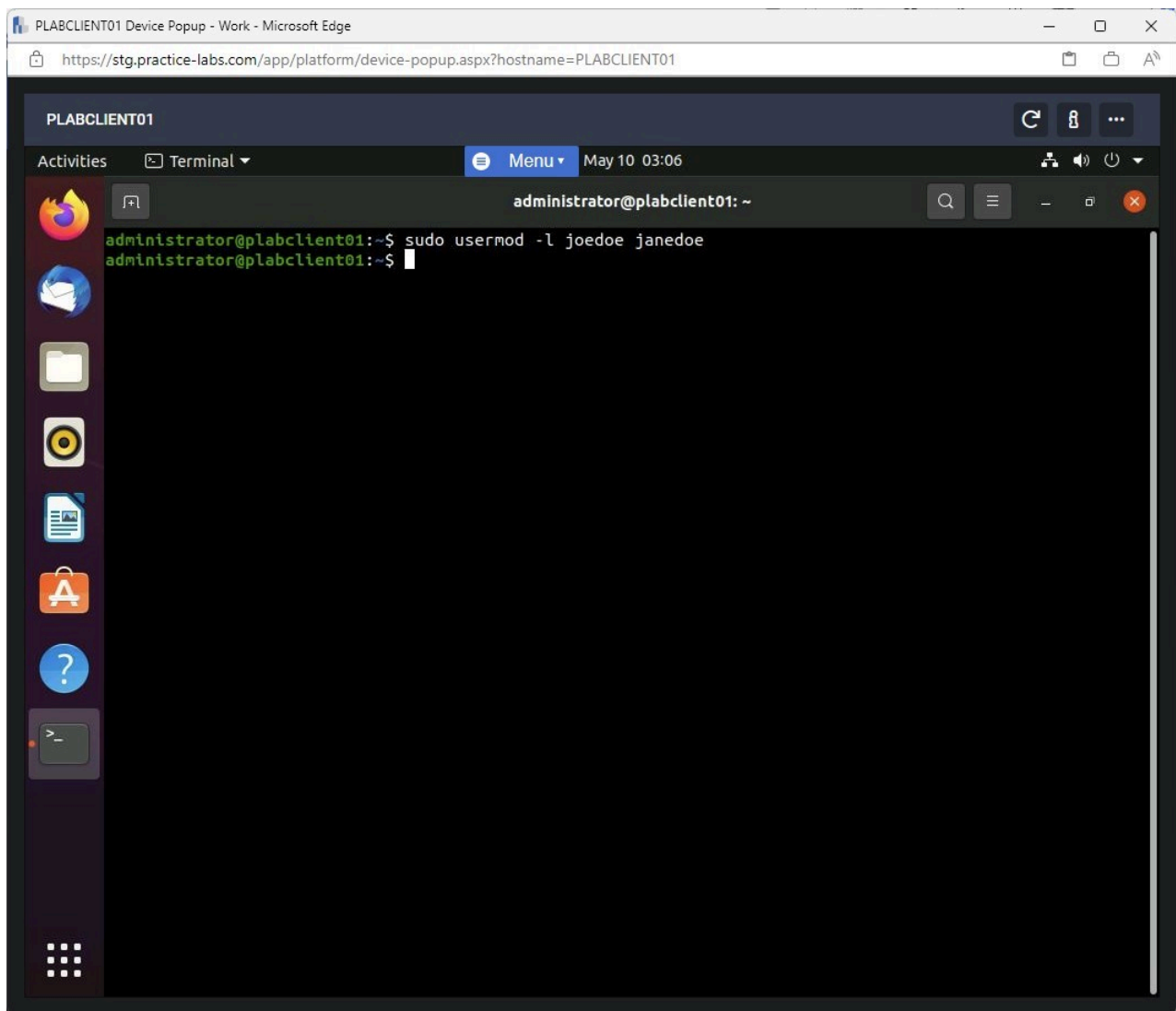


Figure 1.19 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **usermod** command is used to modify the user account. The **-l** option provides the new value for the login name and changes the name of the user account from **janedoe** to **joedoe**.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
sudo less /etc/passwd
```

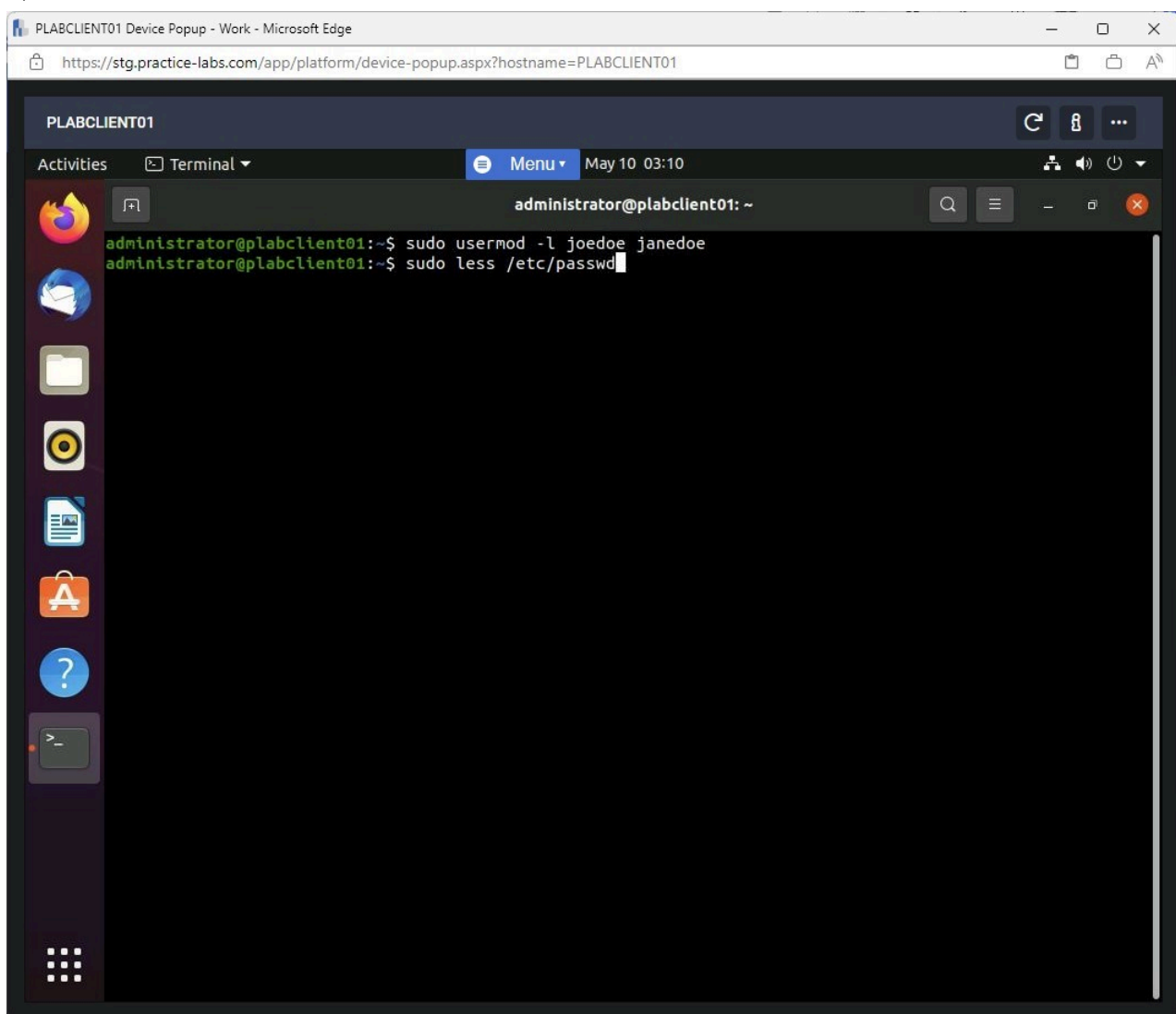



Figure 1.20 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 3

Scroll down the text file. Notice the renamed **joedoe** account is listed.

Type the following to exit out of the window:

q

```

PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 10 03:12
administrator@plabclient01: ~
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106:/nonexistent:/usr/sbin/nologin
syslog:x:104:110:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,:/var/lib/tpm:/bin/false
uidd:x:107:114:/run/uidd:/usr/sbin/nologin
tcpdump:x:108:115:/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:109:116:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:110:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin
rtkit:x:111:117:RealtimeKit,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:113:120:user for cups-pk-helper service,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:114:29:Speech Dispatcher,,:/run/speech-dispatcher:/bin/false
avahi:x:115:121:Avahi mDNS daemon,,:/var/run/avahi-daemon:/usr/sbin/nologin
kernoops:x:116:65534:Kernel Oops Tracking Daemon,,:/usr/sbin/nologin
saned:x:117:123:/var/lib/saned:/usr/sbin/nologin
nm-openvpn:x:118:124:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
hplip:x:119:7:HPLIP system user,,:/run/hplip:/bin/false
whoopsie:x:120:125:/nonexistent:/bin/false
colord:x:121:126:colord colour management daemon,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:122:127:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:123:128:PulseAudio daemon,,:/var/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:124:65534:/run/gnome-initial-setup:/bin/false
gdm:x:125:130:Gnome Display Manager:/var/lib/gdm3:/bin/false
sssd:x:126:131:SSSD system user,,:/var/lib/sss:/usr/sbin/nologin
administrator:x:1000:1000:administrator,,:/home/administrator:/bin/bash
systemd-coredump:x:999:999:systemd Core Dumper:/usr/sbin/nologin
joedoe:x:1001:1001:Jane Doe:/home/janedoe:/bin/sh
(END)

```

Figure 1.21 Screenshot of PLABCLIENT01: The /etc/passwd text file is displayed, showing the renamed account.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
sudo usermod -L joedoe
```

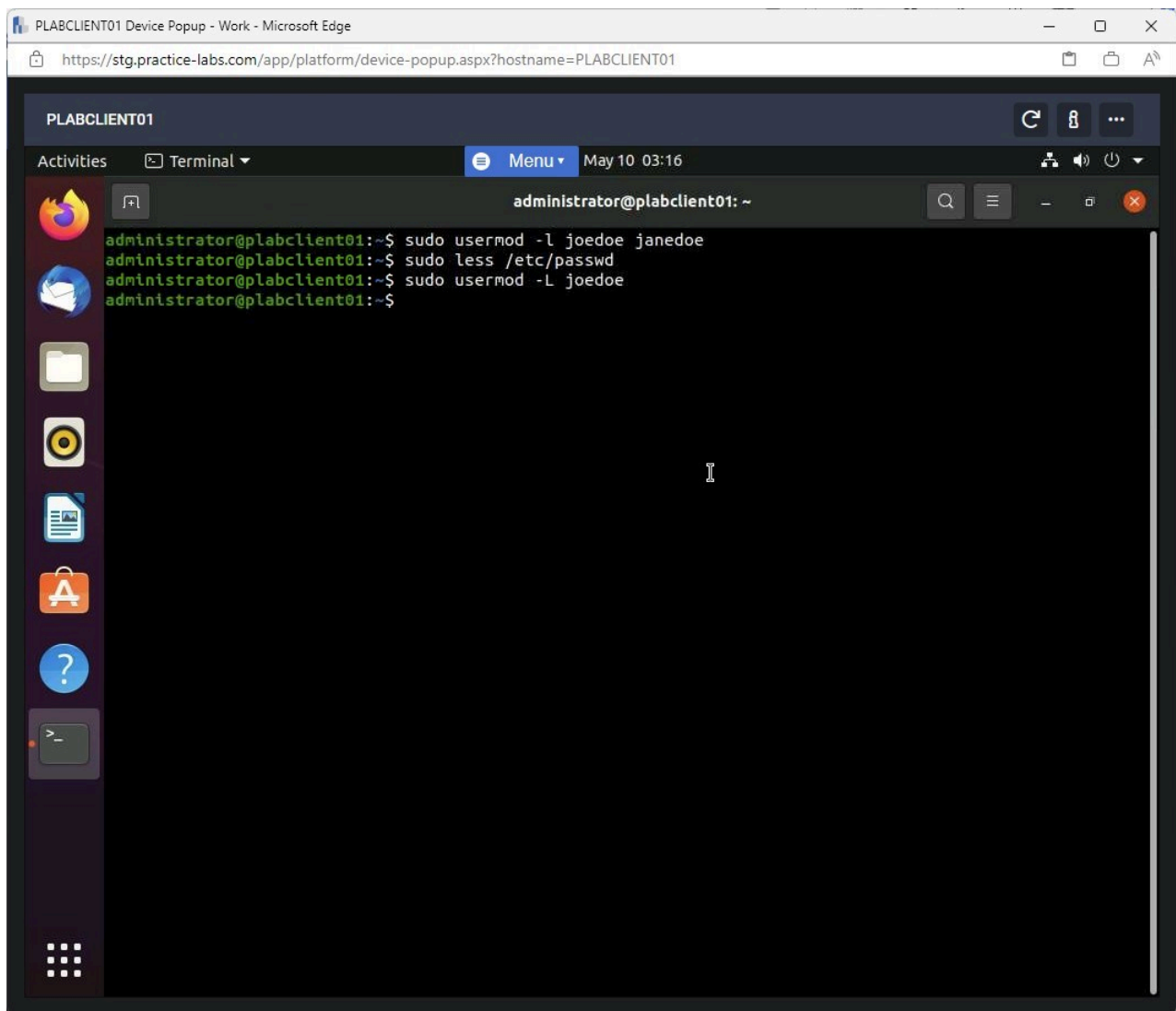



Figure 1.22 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **-L** option will lockout the **joedoe** account and prevents the user from accessing the system.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
sudo usermod -U joedoe
```

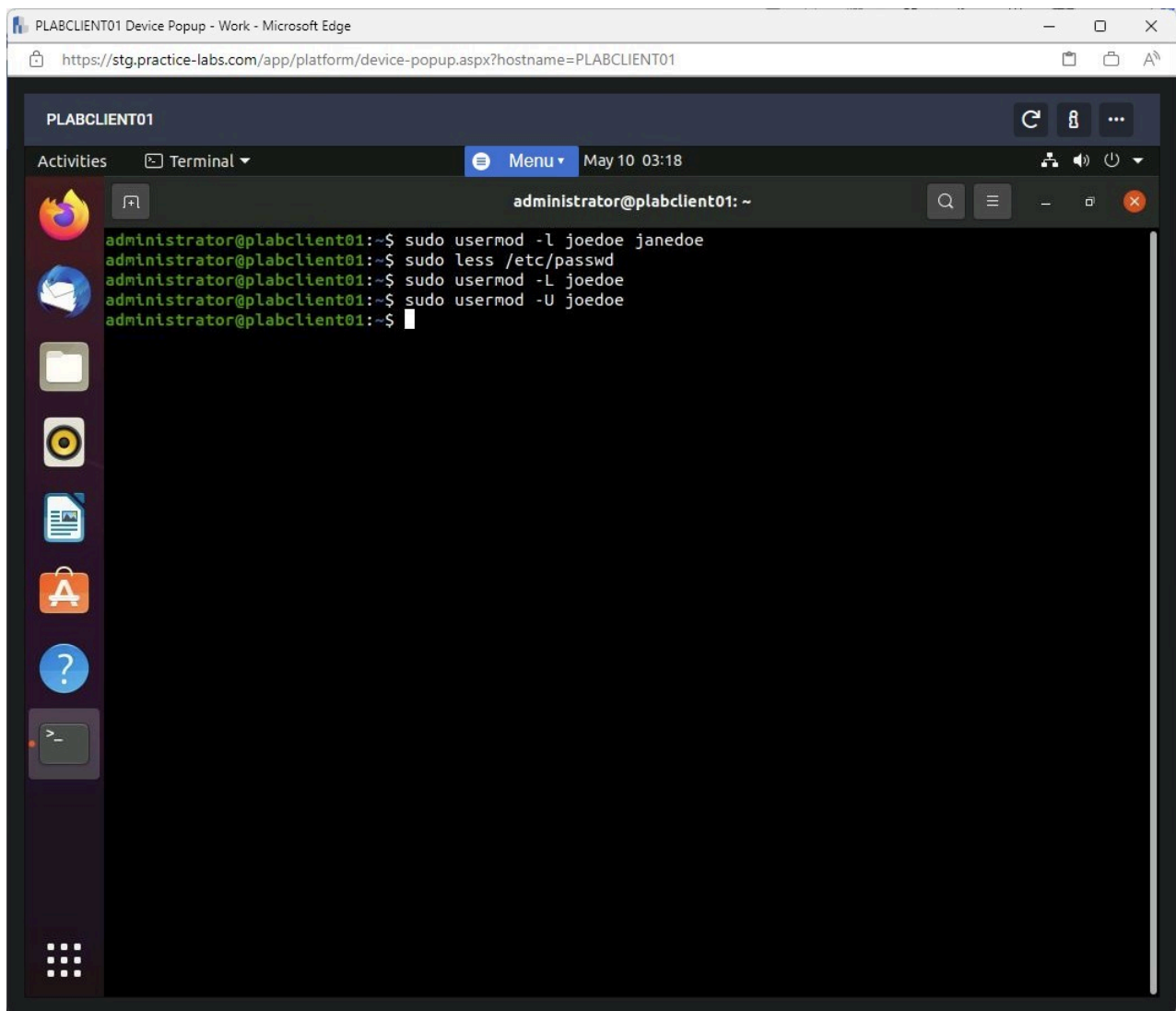


Figure 1.23 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *-U* option will unlock the **joedoe** account.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
sudo chsh -s /bin/login joedoe
```

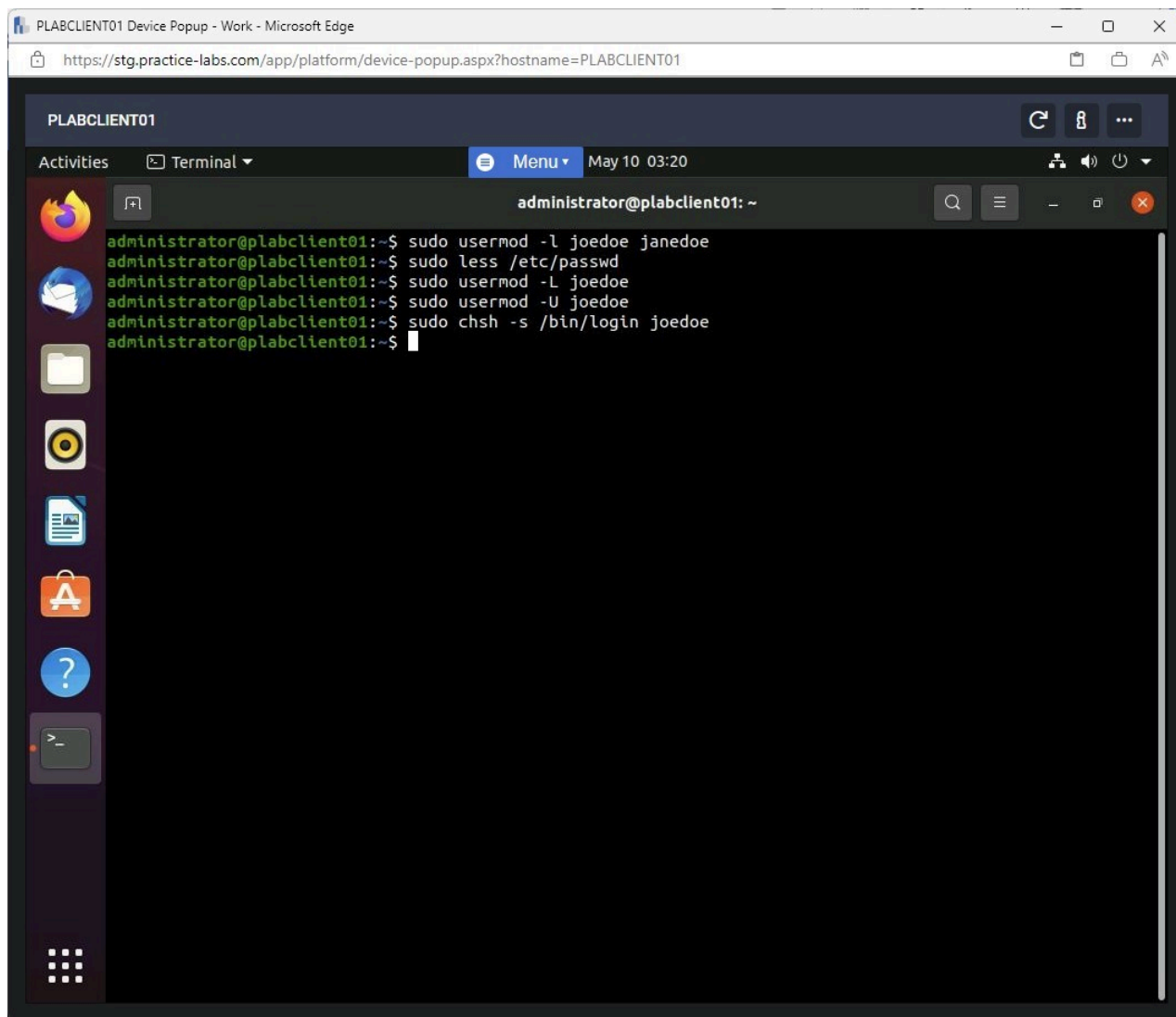


Figure 1.24 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *chsh* (change shell) command will modify the account to become a service account. This prevents the account from logging on interactively but is still useful for running services.

Keep the **Terminal** window open.

Task 3 - Delete the User Account

In this task, you will use the **userdel** command to delete the **joedoe** account.

Step 1

In **PLABCLIENT01**, the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo userdel -r joedoe
```

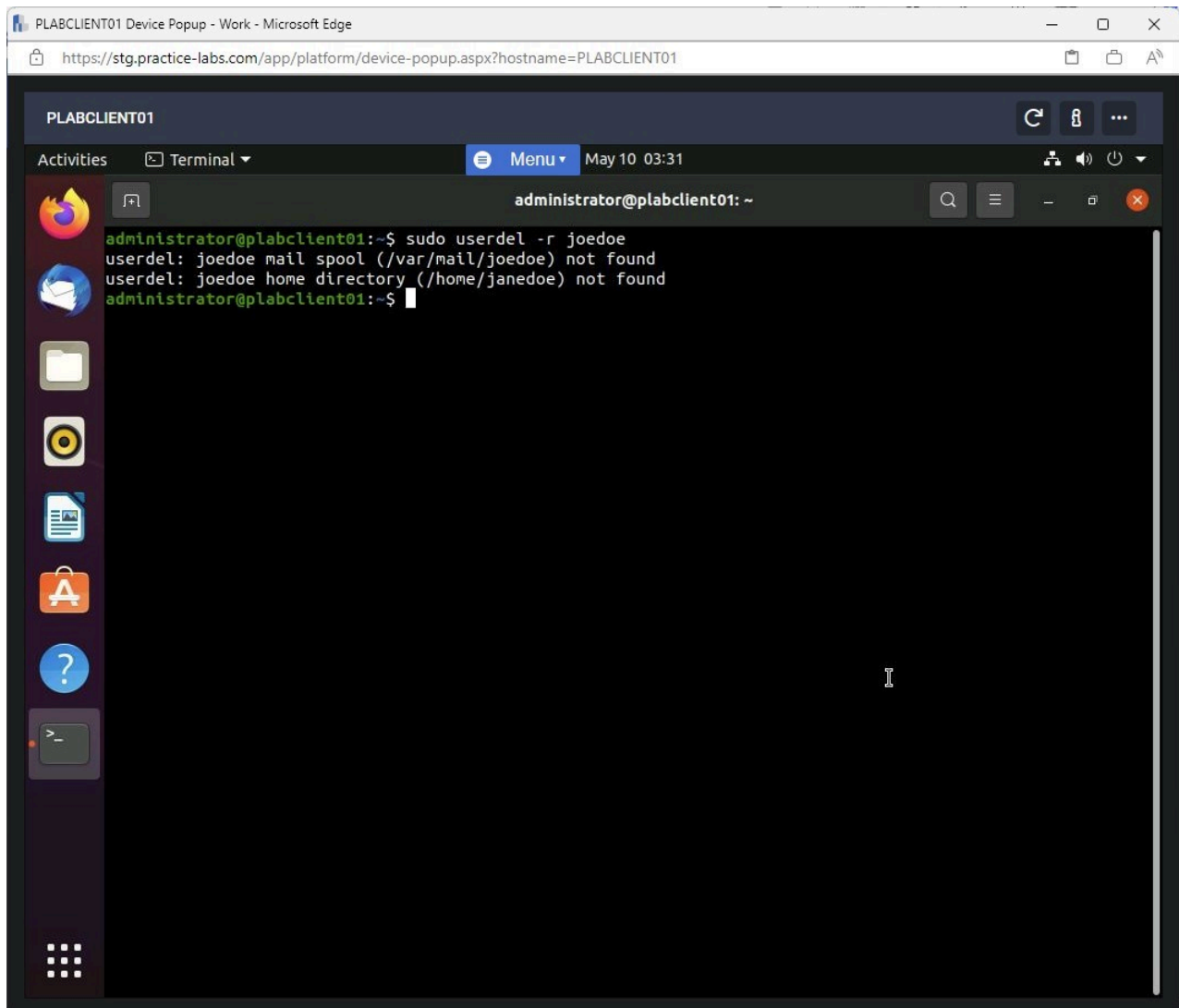


Figure 1.25 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **userdel** command is used to delete the user account. The **-r** option also deletes the user's home folder, that is, the **janedoe** folder. Notice the **not found** output.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
sudo less /etc/passwd
```

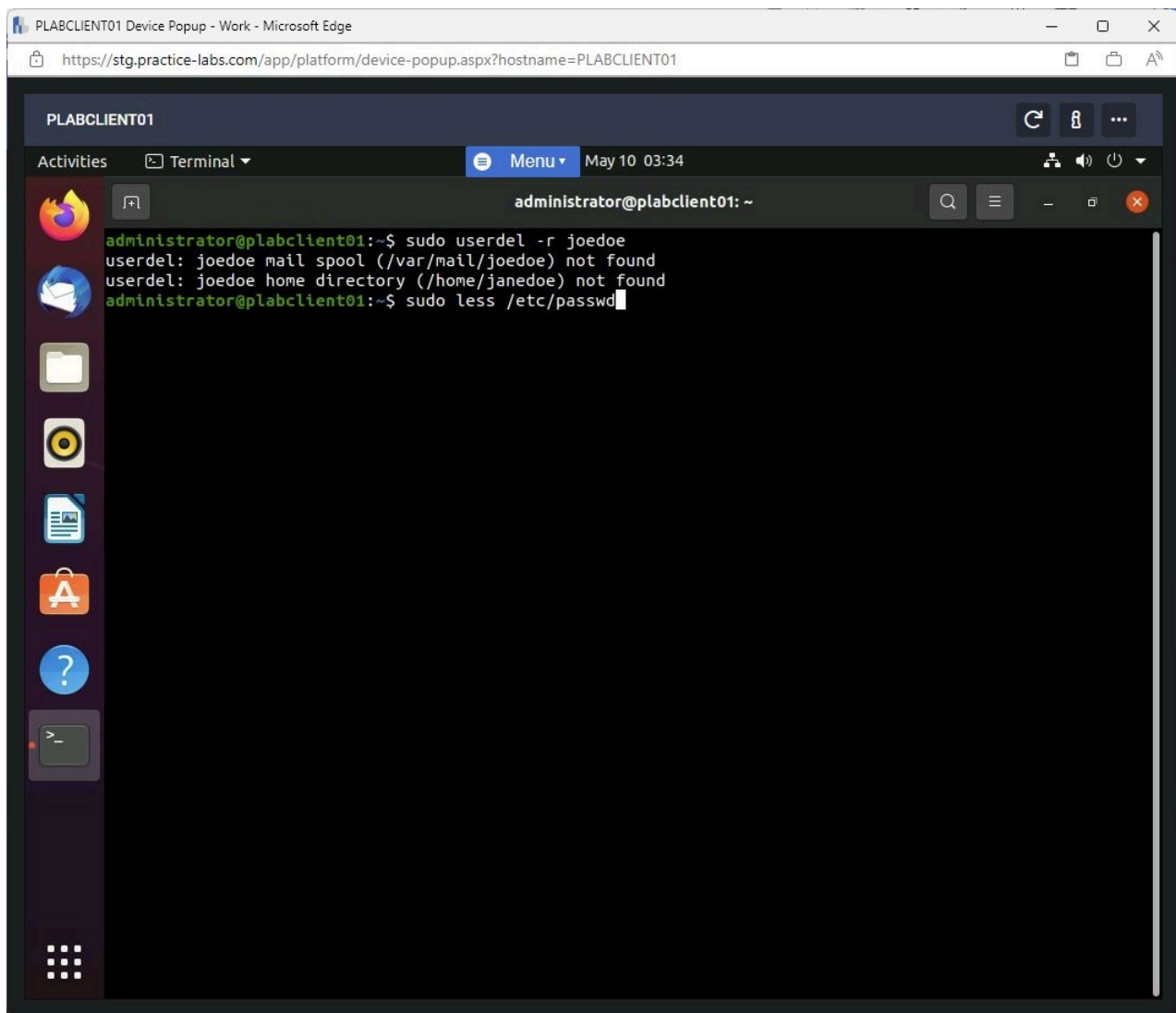


Figure 1.26 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 3

Scroll down the text file. Notice the **joedoe** account doesn't exist anymore.

Type the following to exit out of the window:

q

PLABCLIENT01 Device Popup - Work - Microsoft Edge

https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01

Activities Terminal Menu May 10 03:36

administrator@plabclient01: ~

```
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106:/:nonexistent:/usr/sbin/nologin
syslog:x:104:110:/:home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/:nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:114:/:run/uuidd:/usr/sbin/nologin
tcpdump:x:108:115:/:nonexistent:/usr/sbin/nologin
avahi-autoipd:x:109:116:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:110:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
rtkit:x:111:117:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:113:120:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:114:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
avahi:x:115:121:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/usr/sbin/nologin
kernoops:x:116:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
saned:x:117:123:/:var/lib/saned:/usr/sbin/nologin
nm-openvpn:x:118:124:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
hplip:x:119:7:HPLIP system user,,,:/run/hplip:/bin/false
whoopsie:x:120:125:/:nonexistent:/bin/false
colord:x:121:126:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:122:127:/:var/lib/geoclue:/usr/sbin/nologin
pulse:x:123:128:PulseAudio daemon,,,:/var/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:124:65534:/:run/gnome-initial-setup:/bin/false
gdm:x:125:130:Gnome Display Manager:/var/lib/gdm3:/bin/false
sssd:x:126:131:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
administrator:x:1000:1000:administrator,,,:/home/administrator:/bin/bash
systemd-coredump:x:999:999:systemd Core Dumper:/usr/sbin/nologin
(END)
```

Figure 1.27 Screenshot of PLABCLIENT01: The /etc/passwd text file is displayed.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```

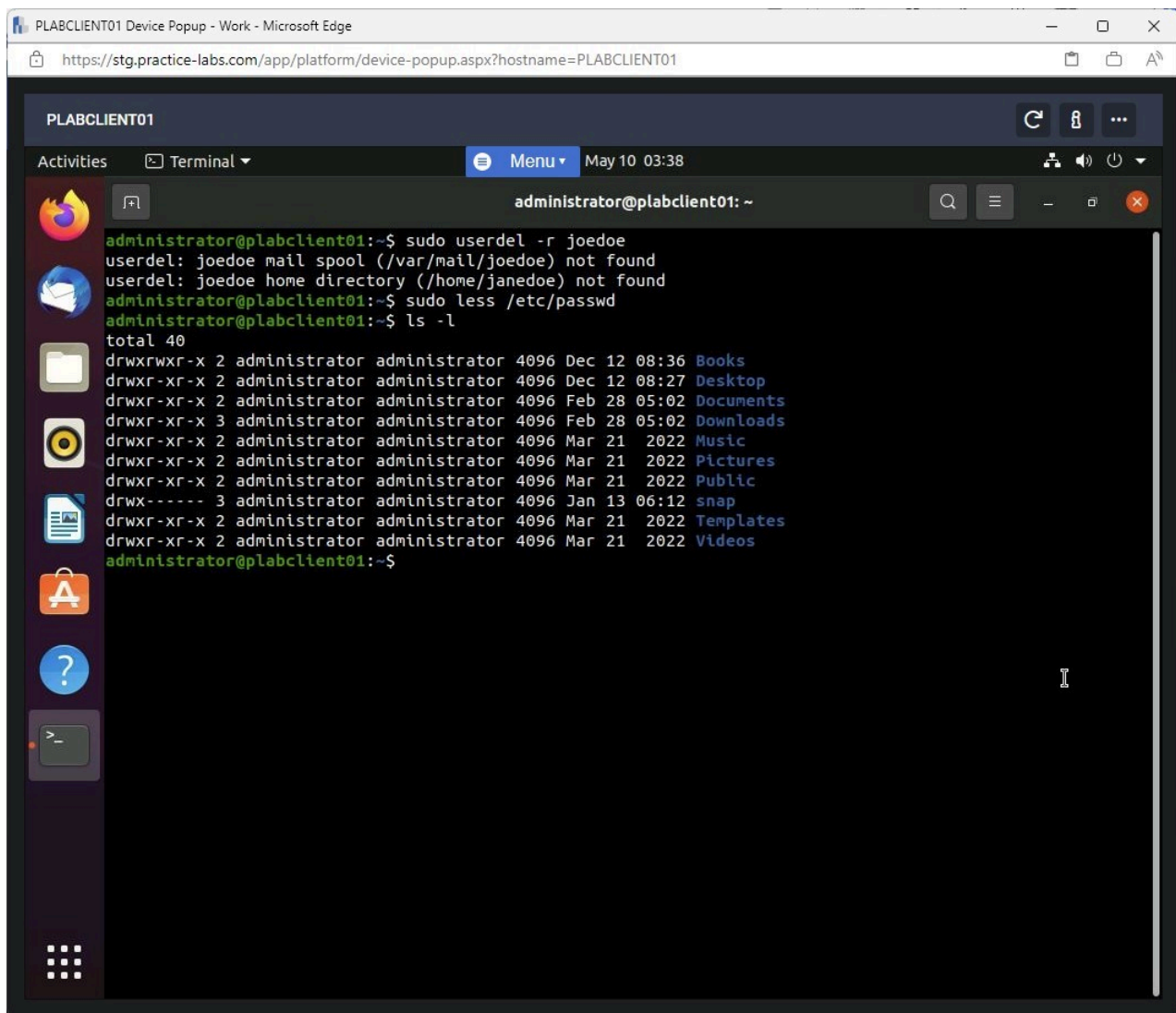



Figure 1.28 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the home directory for **janedoe** has also been deleted.

You can also delete the joedoe account without deleting the home folder by not adding the **-r** option as follows:
sudo userdel joedoe

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
clear
```

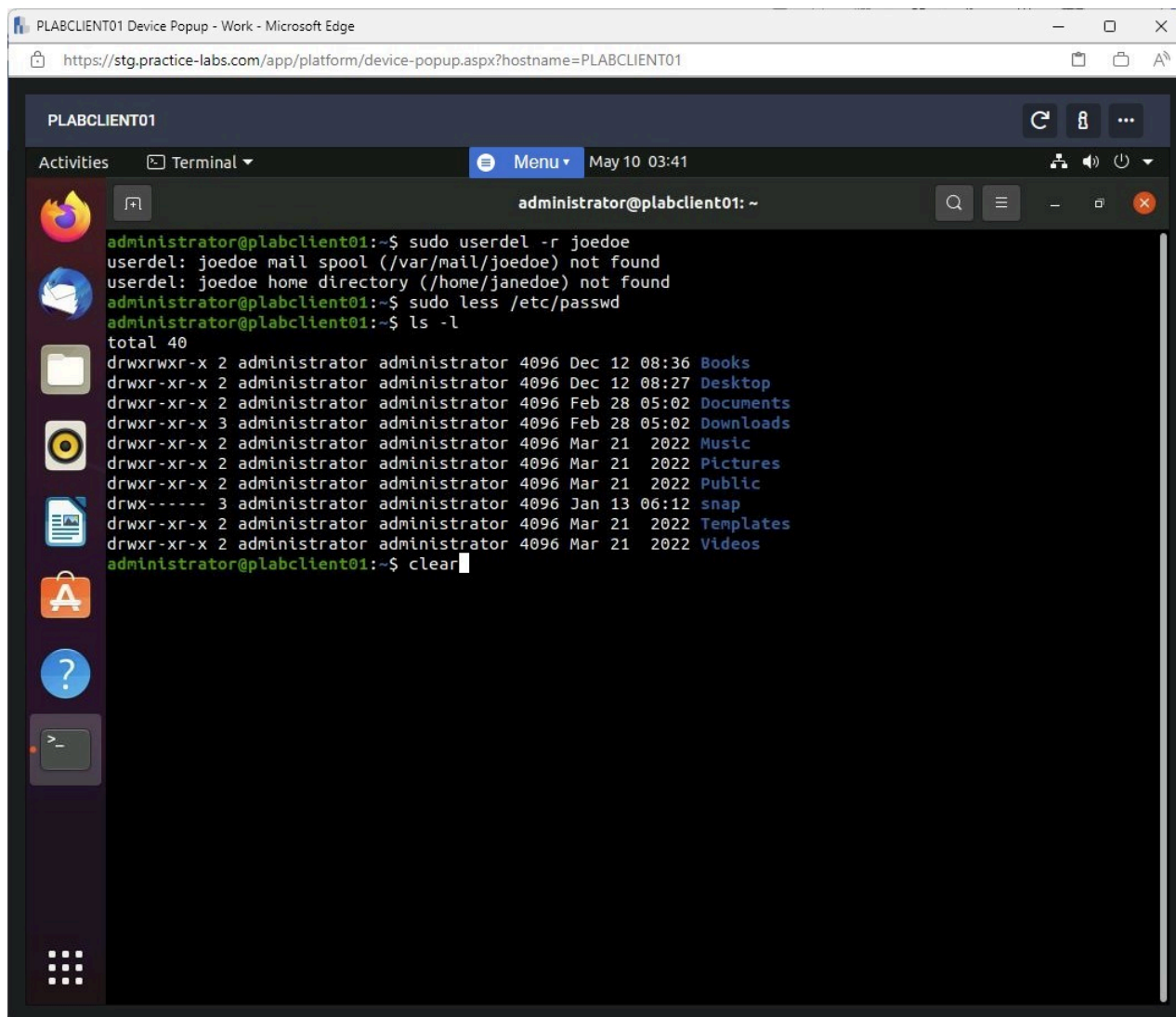



Figure 1.29 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Keep the **Terminal** window open.

Exercise 2 - Create and Configure Groups in Linux

In this exercise, you will create, modify, and add users to groups using the terminal interface.

Learning Outcomes

After completing this exercise, you should be able to:

- Create Groups
- Add Users to Groups

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Create Groups

In this task, you will create groups using the **groupadd** command.

Step 1

Ensure you are connected to **PLABCLIENT01** and the **Terminal** window is open.

In the **Terminal** window, type the following commands to add three groups (Press **Enter** after each command):

```
sudo groupadd Marketing  
sudo groupadd Sales  
sudo groupadd IT
```

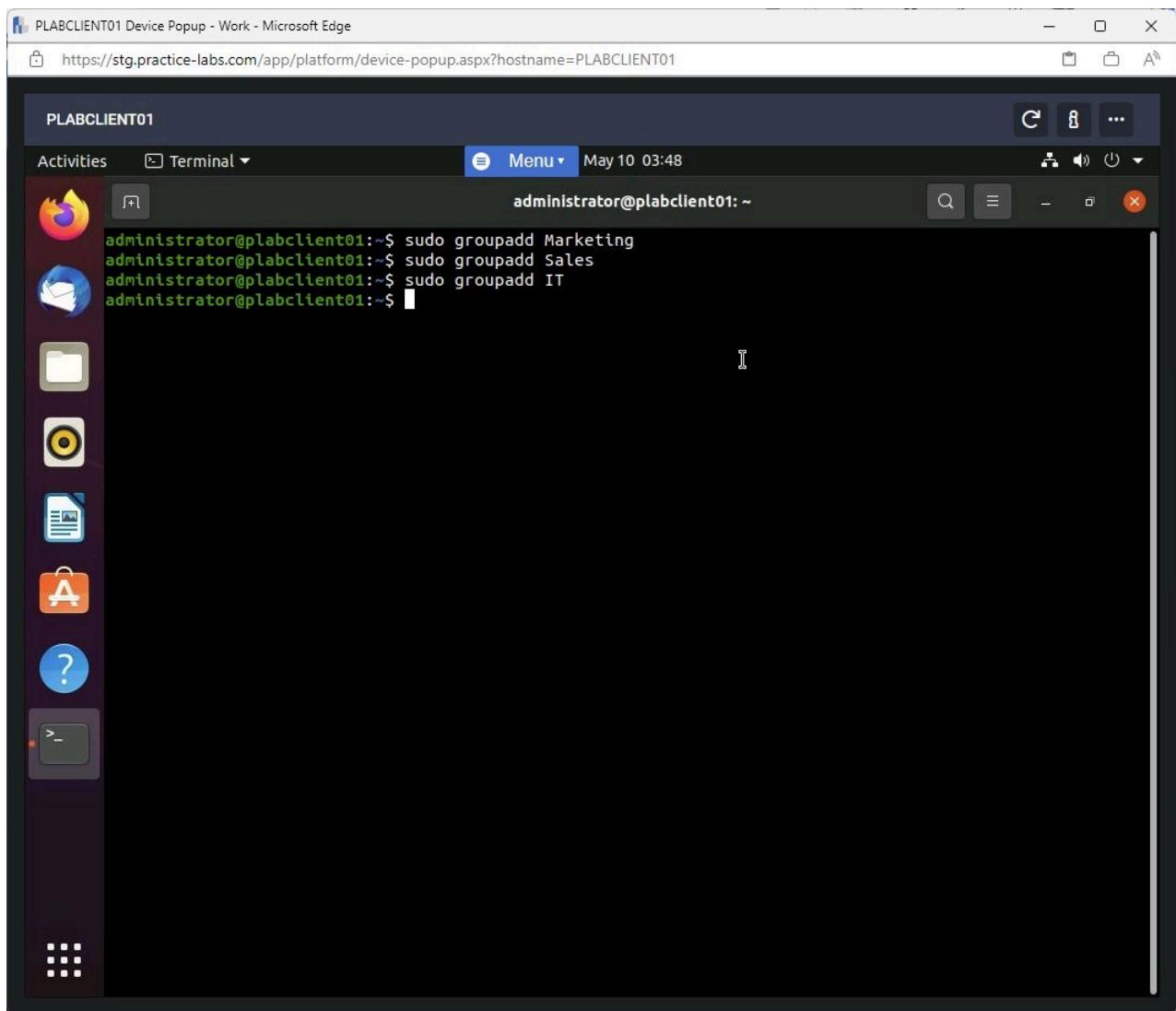


Figure 2.1 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
sudo tail /etc/group
```

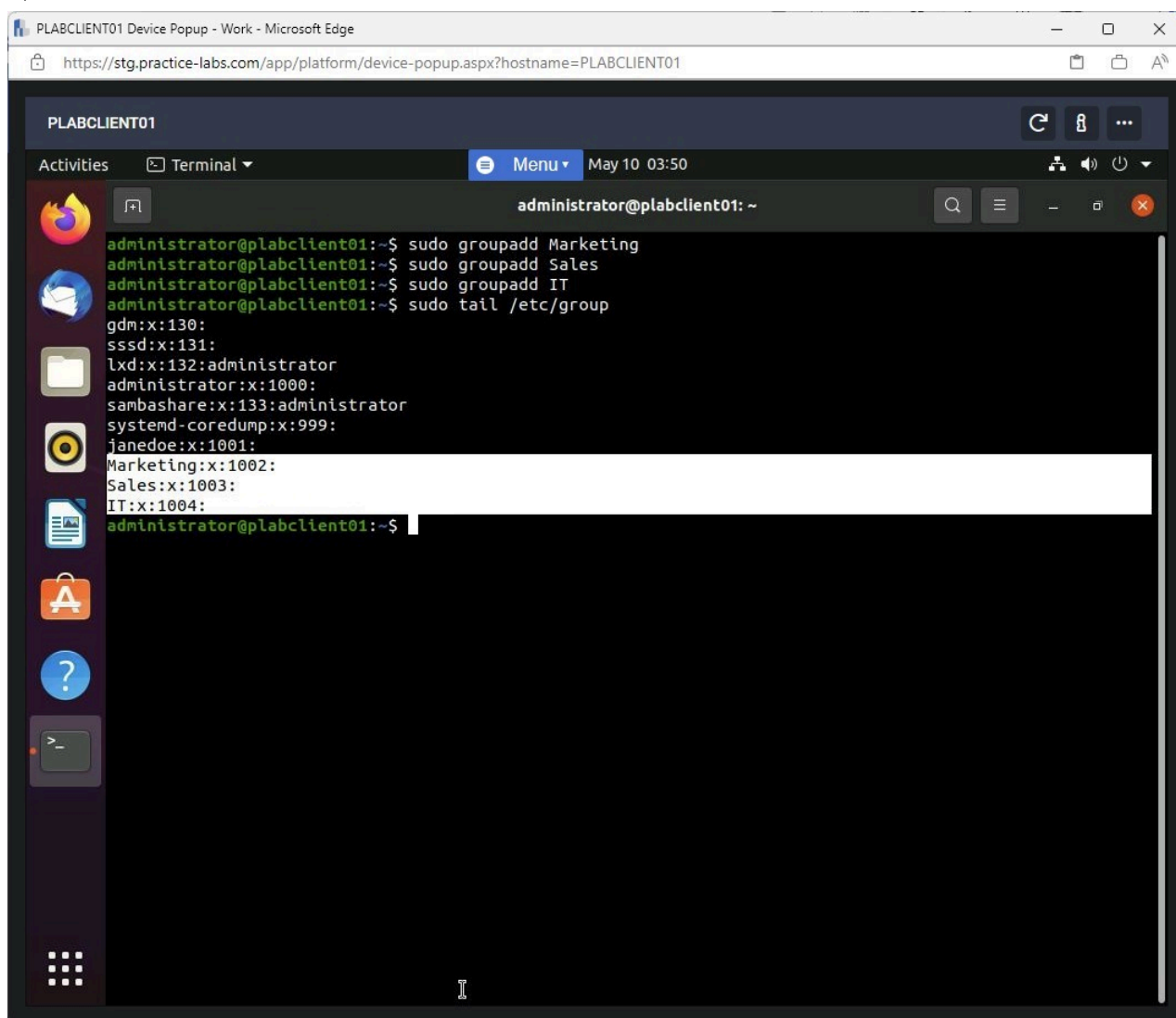


Figure 2.2 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the three groups are successfully created.

Keep the **Terminal** window open.

Task 2 - Add Users to Groups

In this task, you will create new users and then use the **gpaswd** command to add the users to the groups.

Step 1

In **PLABCLIENT01**, the **Terminal** window is open.

In the **Terminal** window, type the following commands to add three new users (Press **Enter** after each command):

```
sudo useradd jimdoe  
sudo useradd juliedoe  
sudo useradd jackdoe
```

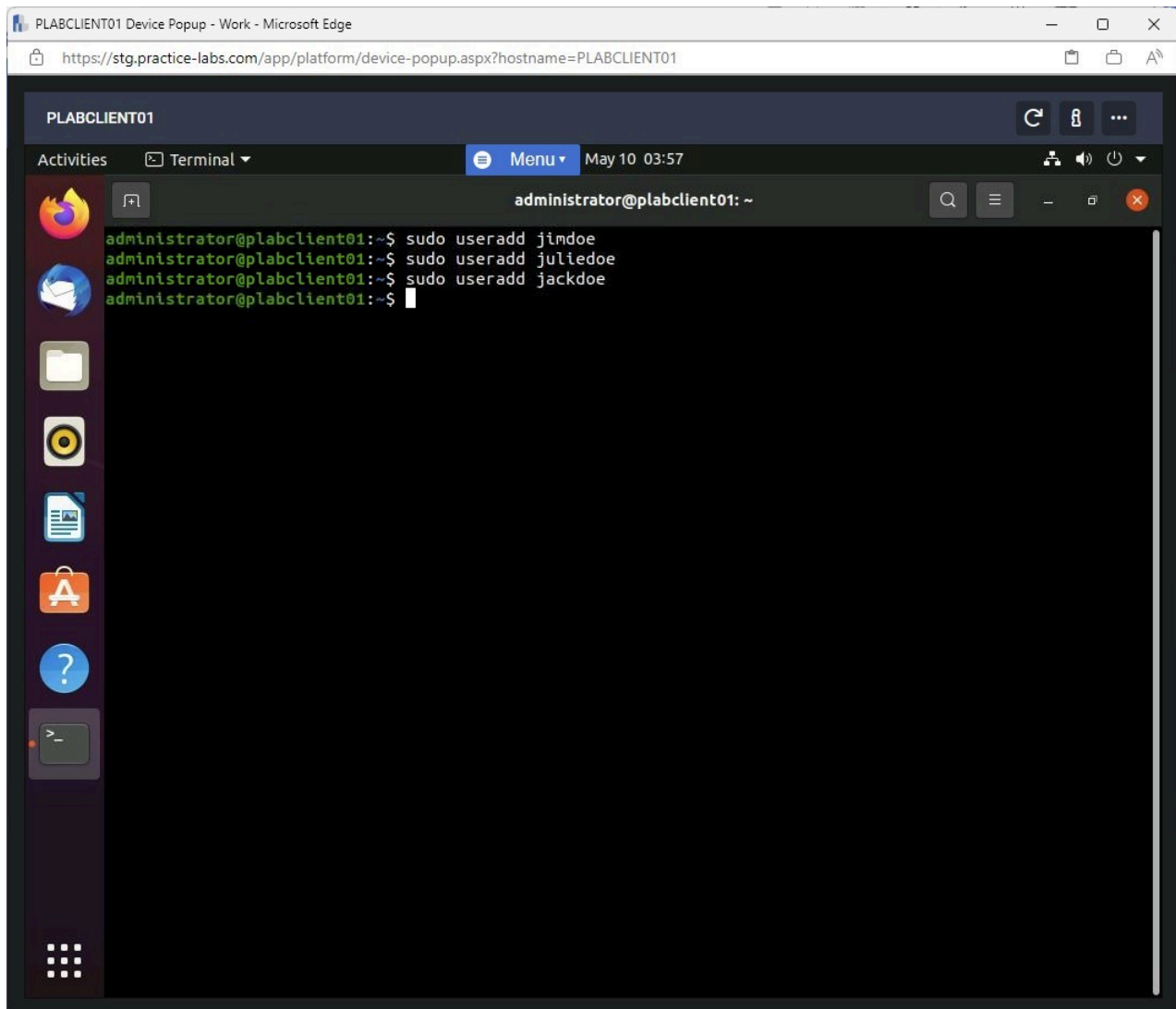


Figure 2.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Step 2

In the **Terminal** window, type the following commands to add the new users to the three groups that were created earlier (Press **Enter** after each command):

```
sudo gpasswd -a jimdoe Marketing  
sudo gpasswd -a juliedoe Sales
```

```
sudo gpasswd -a jackdoe IT
```

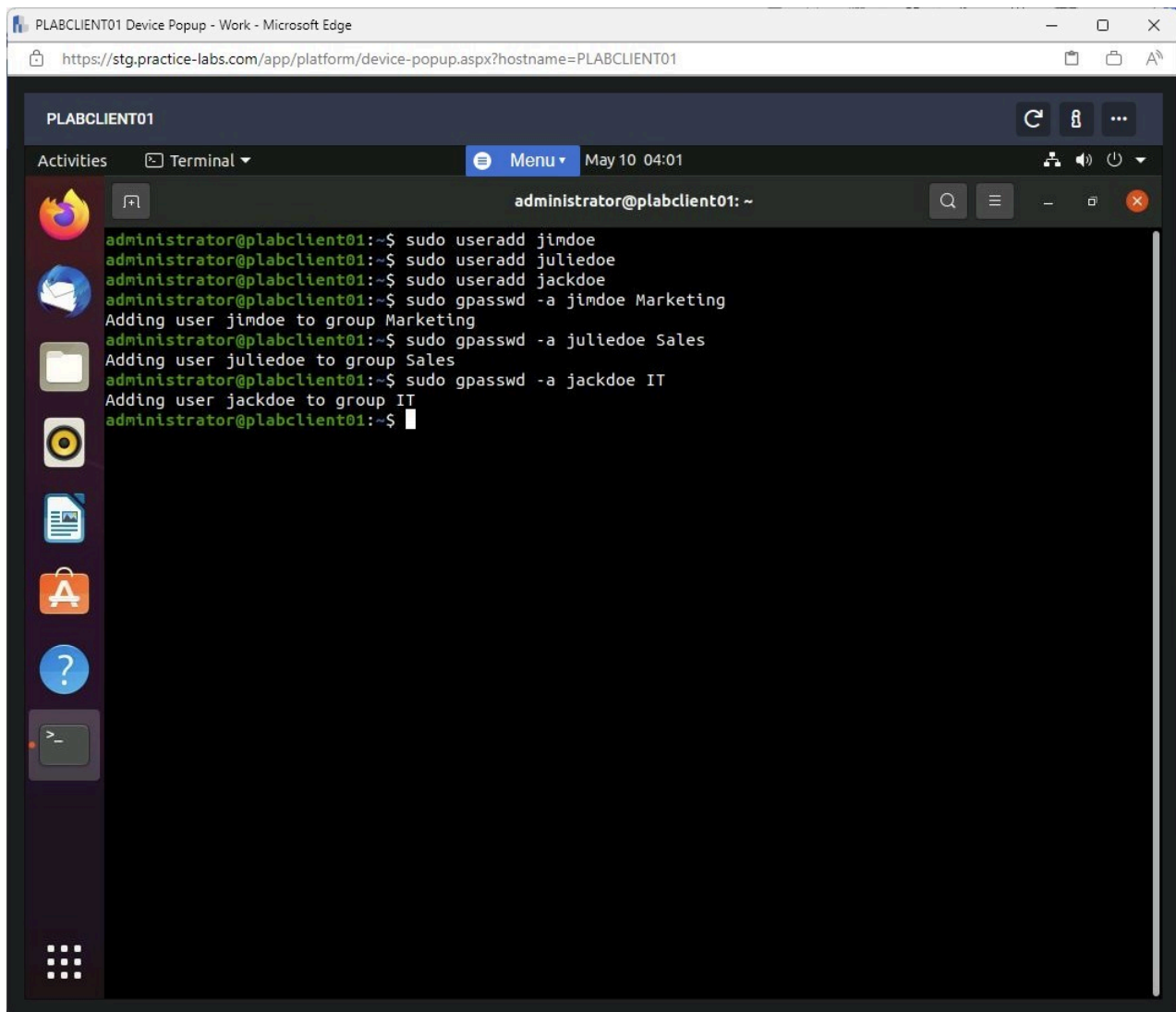


Figure 2.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The three new users are now successfully added to the respective groups.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
sudo gpasswd -A juliedoe Sales
```

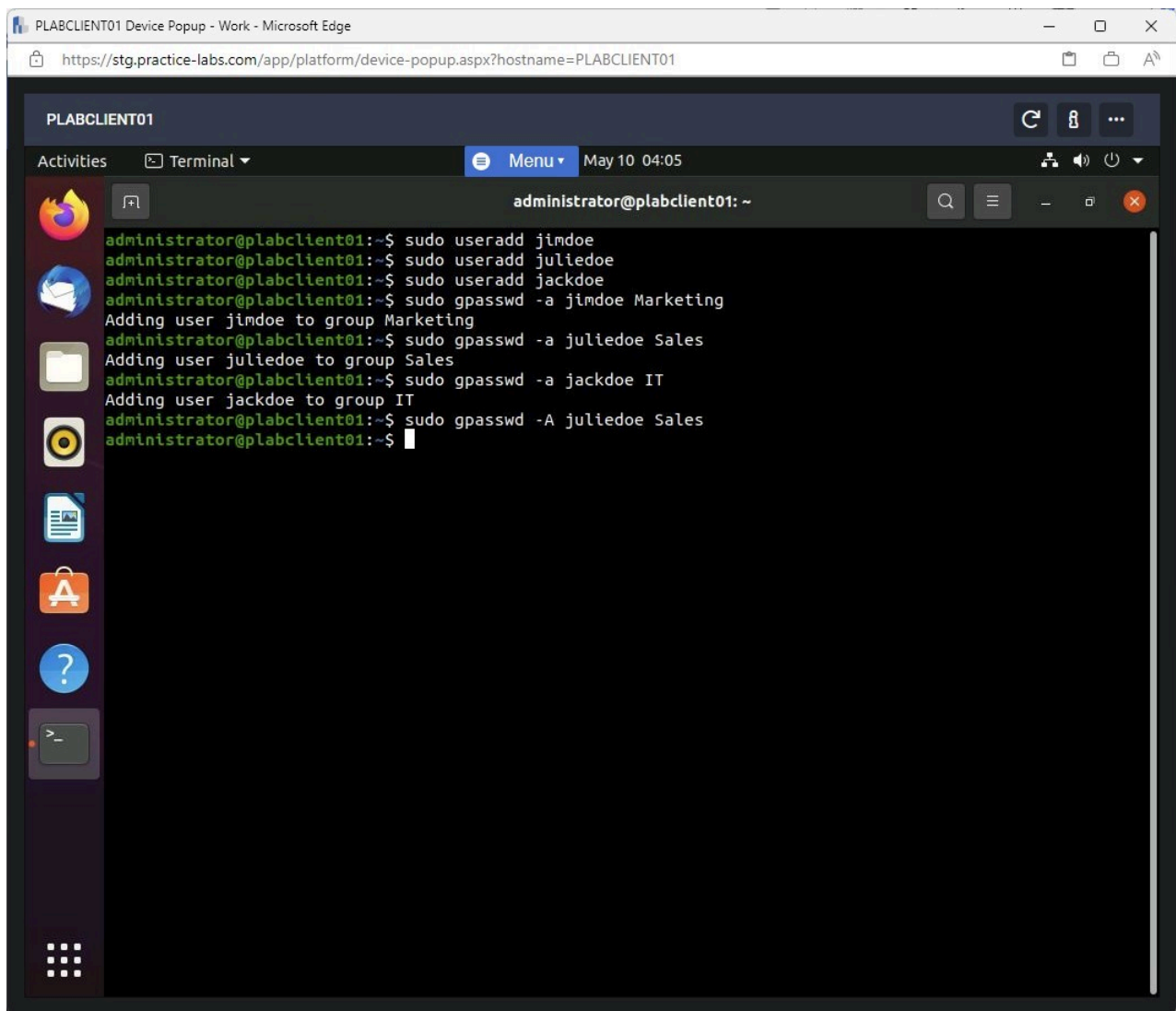


Figure 2.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **-A** option will make **juliedoe** an admin user in the group.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
sudo gpasswd -d jimdoe Marketing
```

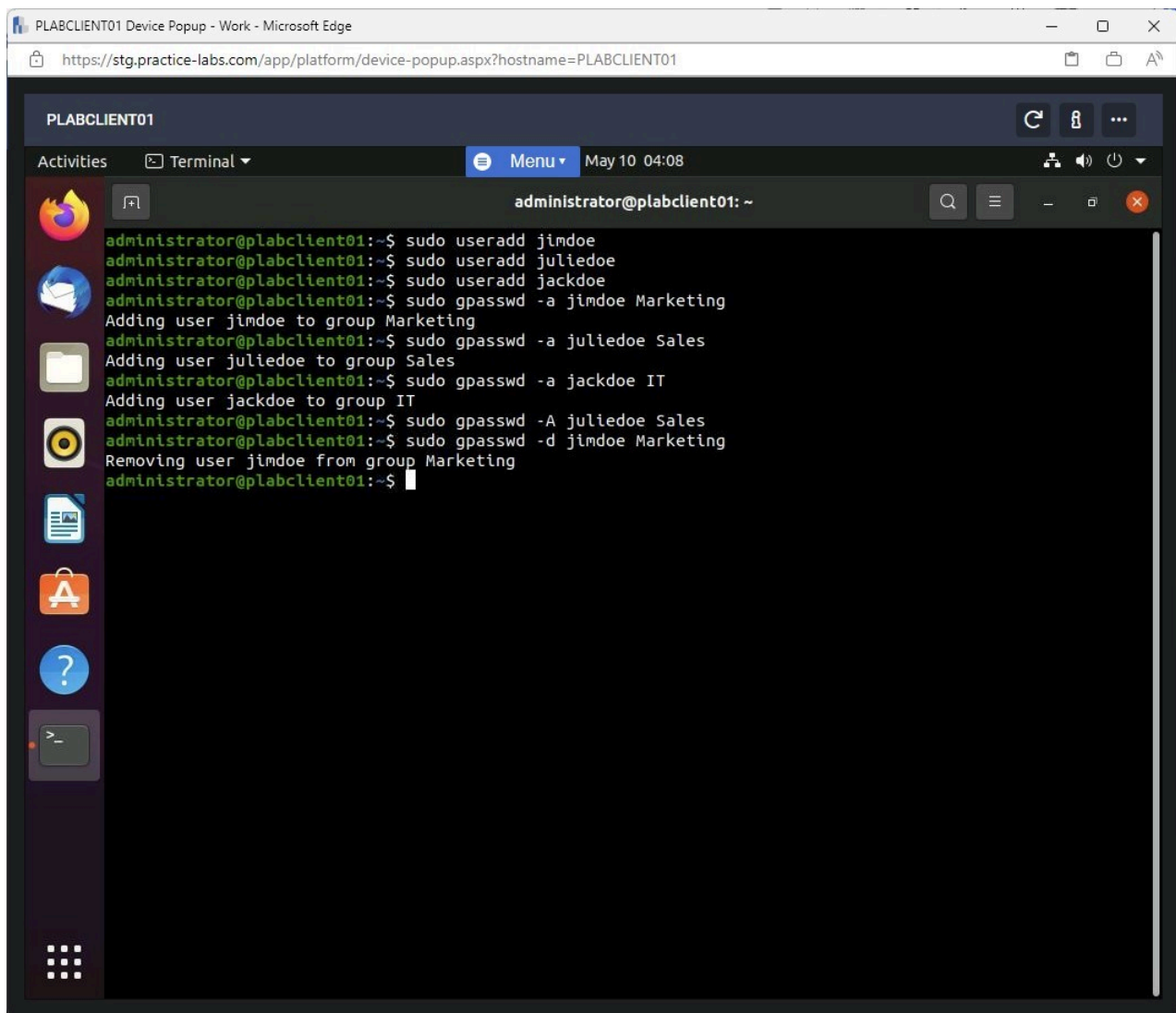



Figure 2.6 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

***Note:** The **-d** option will remove **jimdoe** from the **Marketing** group.*

Step 5

Next, you will have a user run a command with a different group.

In the **Terminal** window, type the following command to navigate to the **Documents** folder:

```
cd Documents/
```

Press **Enter**.

Next, type the following command and press **Enter**:

```
touch file1.txt
```

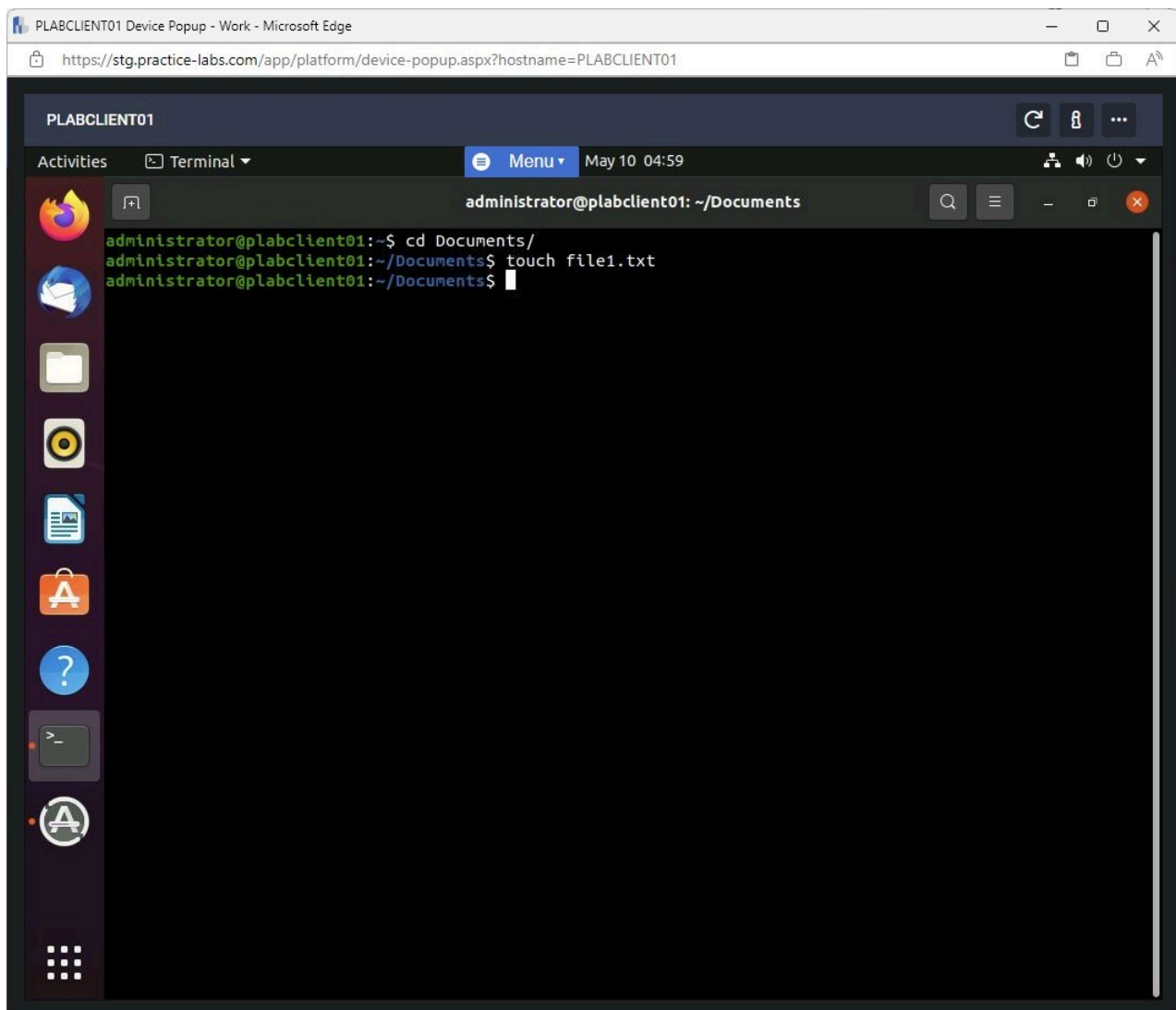


Figure 2.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **touch** command is used to create a new text file.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```

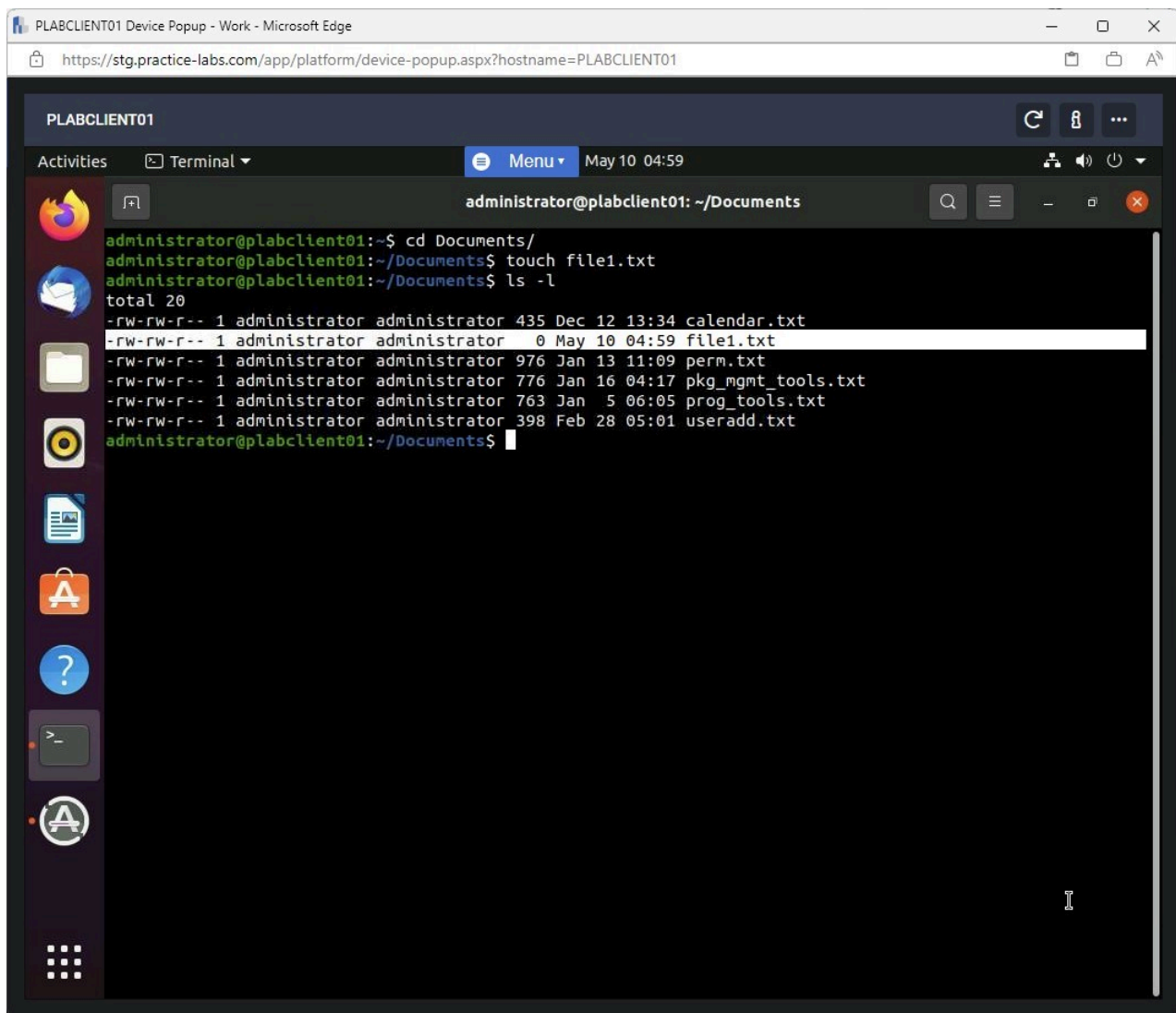


Figure 2.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the user and the group are both **administrator** for **file1.txt**.

Step 7

In the **Terminal** window, type the following command and press **Enter**:

```
newgrp adm
```

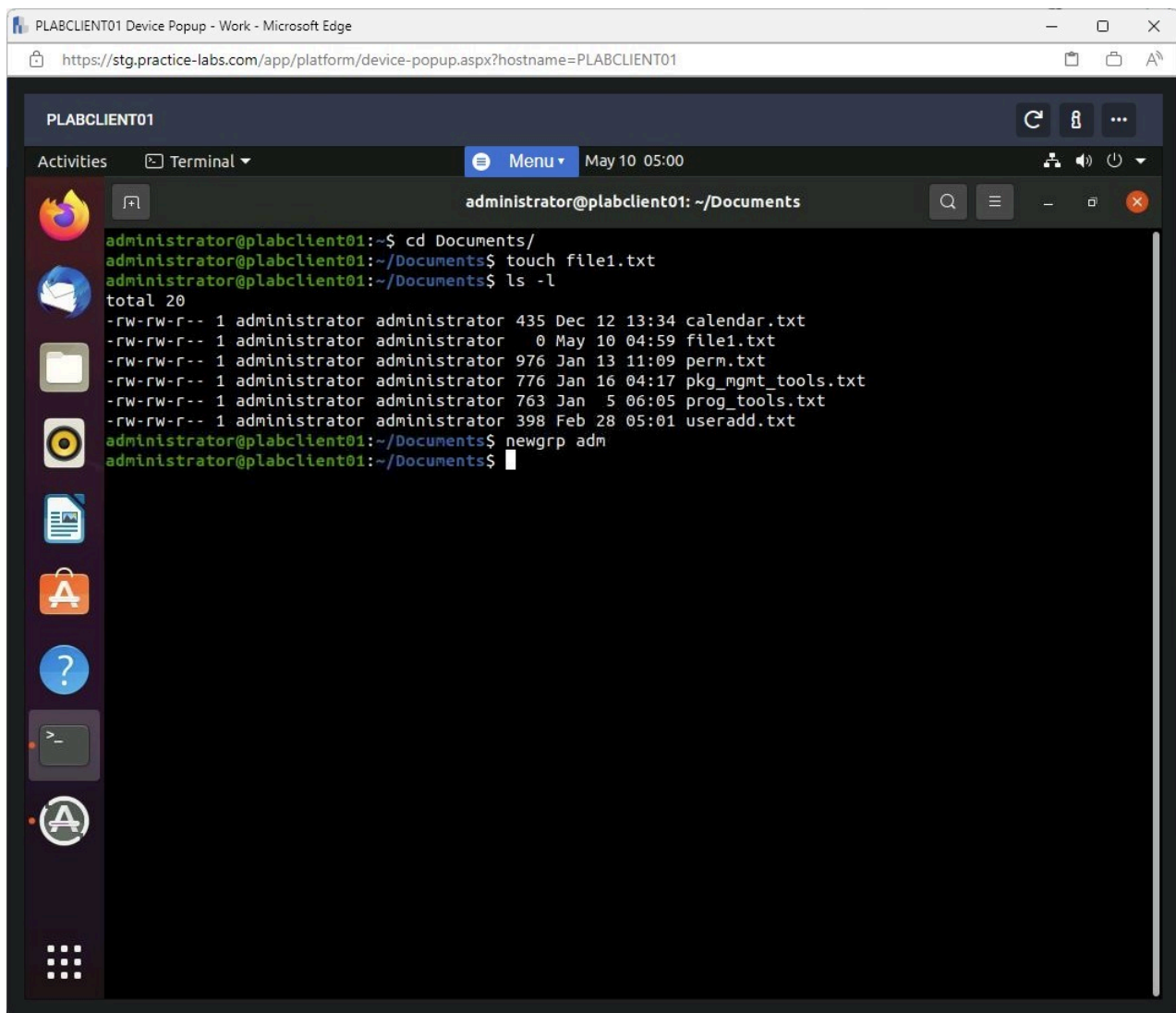


Figure 2.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **newgrp** command changes the name of the user's real group to the group name specified, that is, **adm**.

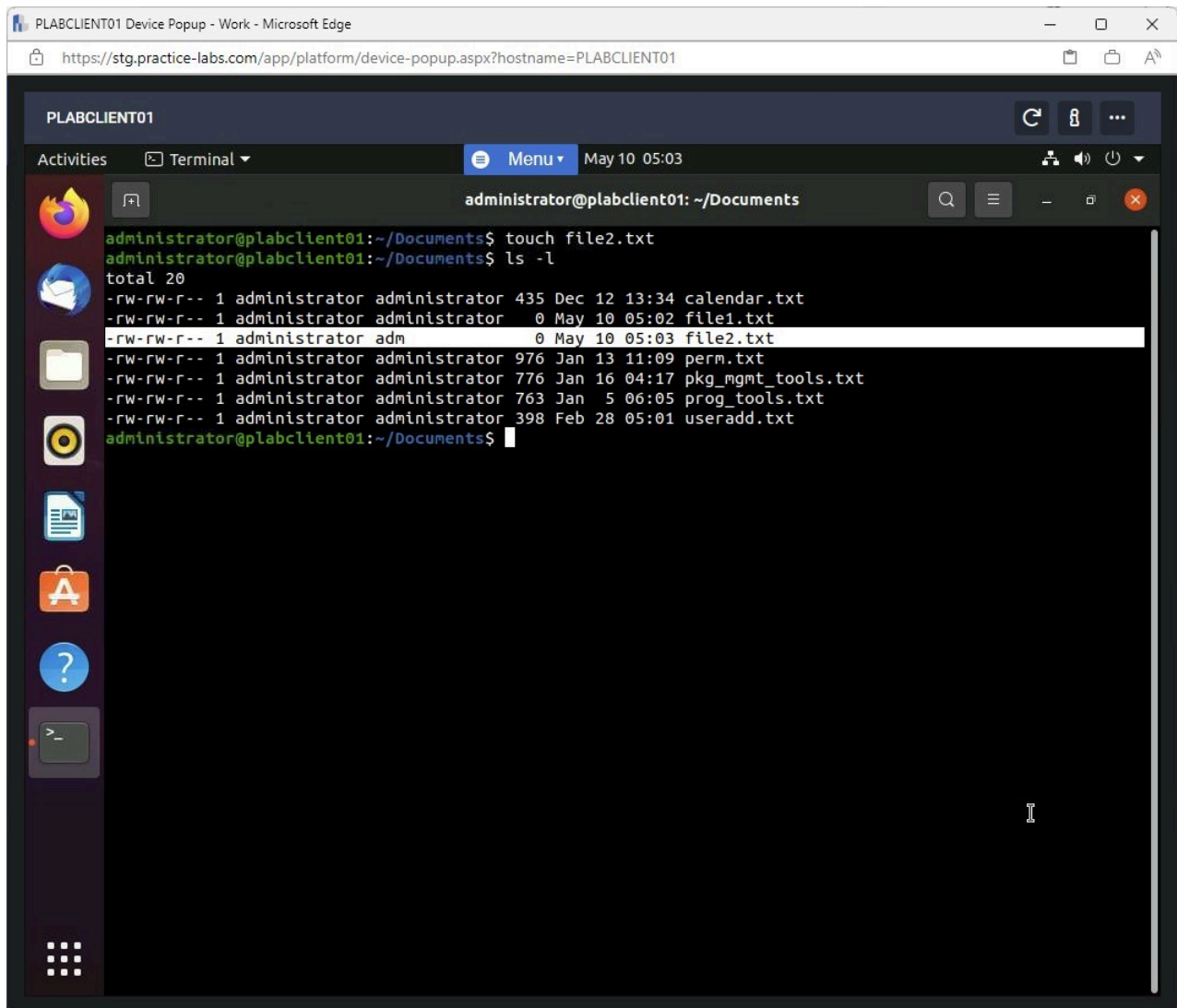
Step 8

In the **Terminal** window, type the following command and press **Enter**:

```
touch file2.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 10 05:03
administrator@plabclient01: ~/Documents
administrator@plabclient01:~/Documents$ touch file2.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 0 May 10 05:02 file1.txt
-rw-rw-r-- 1 administrator adm 0 May 10 05:03 file2.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 2.10 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: Notice the group for *file2.txt* is now *adm*.

Step 9

In the **Terminal** window, type the following command and press **Enter**:

```
getent group
```

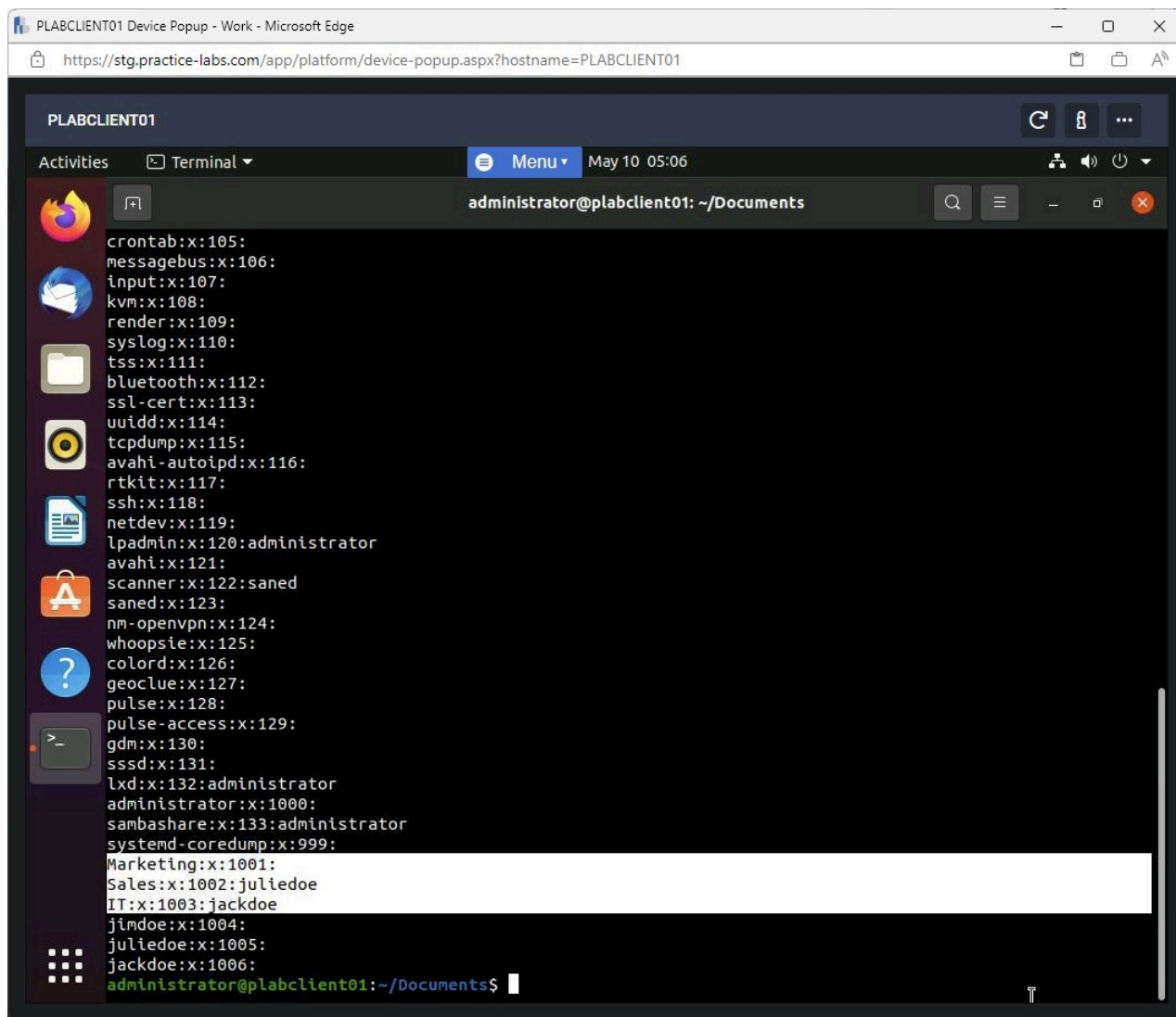


Figure 2.11 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *getent* command is used to verify the users and their groups. Notice the groups and the users in the group that were created earlier.

Step 10

In the **Terminal** window, type the following command and press **Enter**:

```
getent passwd juliedoe
```

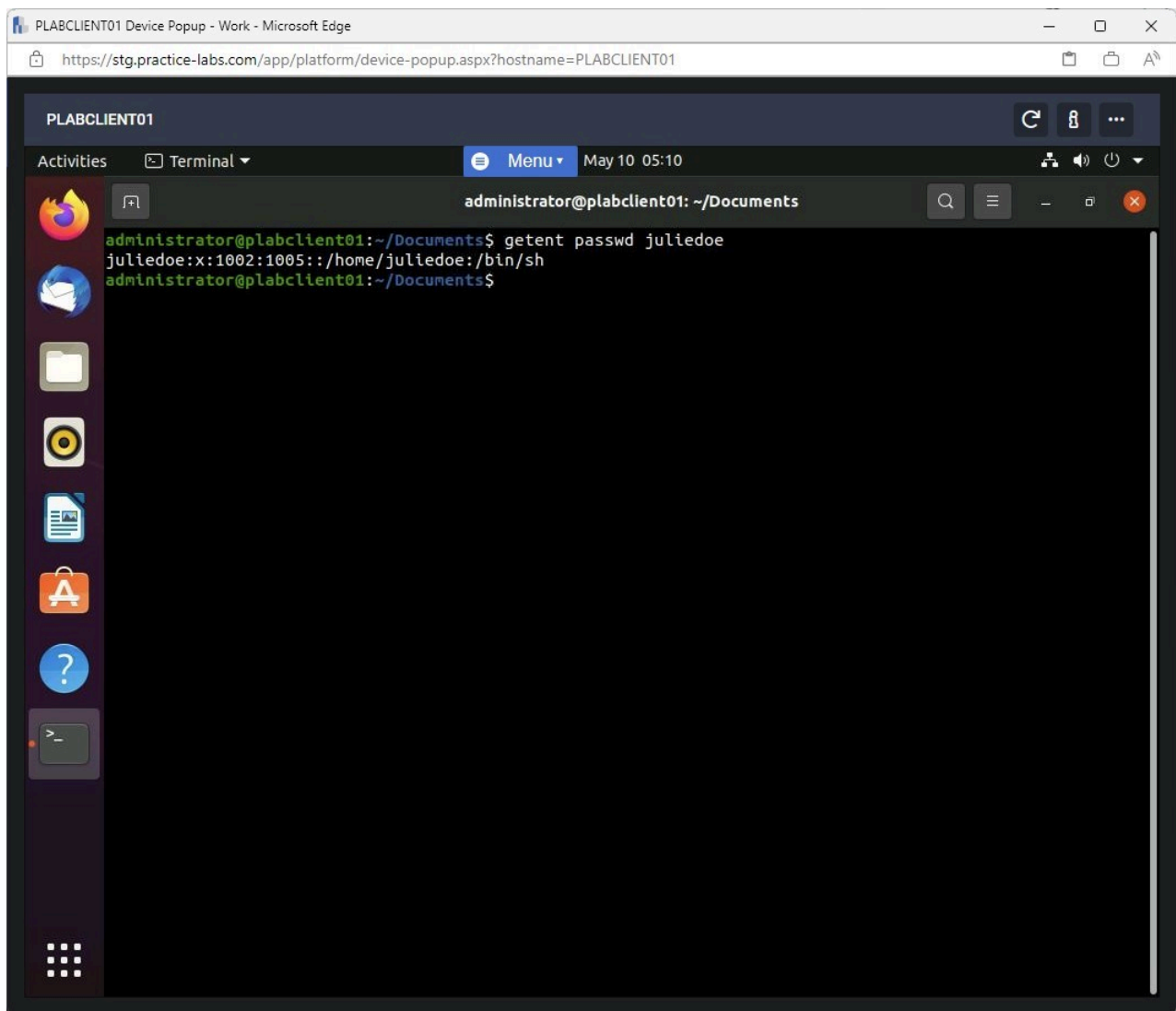



Figure 2.12 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The above *getent* command will display the details for the user *juliedoe*.

Close the **Terminal** window.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Linux Identity Management** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Create and Configure User Accounts in Linux
- Exercise 2 - Create and Configure Groups in Linux

You should now be able to:

- Create a User Account
- Modify the User Account
- Delete the User Account
- Create Groups
- Add Users to Groups

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.